A Dissertation Final Report on
Pet Buy & Sell Website

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Abstract

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In a time when technology is smoothly incorporated into daily life, my project seeks to provide a user-centric solution to link pet lovers with their cherished pets and necessary pet supplies. This paper summarises the process and results of our effort to plan, create, and implement a pet supply website. The pet store website project includes a wide range of features, such as user authentication, easy pet and product searching, dynamic pet/product/category management, user account administration, order tracking and management, thorough order information, and safe payment processing. The main goal of the project, presuming all functionalities are in flawless functioning order, was to guarantee a seamless, simple, and gratifying user experience. Usability testing confirms that the user-centred approach is effective. Users can easily access their accounts thanks to the login process, and the pet and product search tools make it easy for them to find the pets or things they're looking for. It is simple to add and manage pets, items, and categories, enabling effective content management. Strong user and order management capabilities give administrators effective tools for effective management. The payment system provides a safe and convenient checkout process, and order information is given publicly to encourage user trust. Clarity, logical navigation, and visual appeal are prioritised in the website's user interface design to provide a seamless fusion of form and function. Users of all skill levels find the platform to be inclusive and accessible.

This report serves as a testament to our project's path, goals, and accomplishments, emphasising our commitment to providing an outstanding online experience for pet lovers and demonstrating our contribution to the changing e-commerce industry.

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I would like to thank Mr. Khem Raj Sharma, my client, for giving me the responsibility of creating this website to purchase and sell pets. His suggestions, criticism, and cooperation were extremely helpful in making sure the project's goals were realized.

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List of Abbreviations

Short Forms	**Meaning**
E.g.,	Example
etc	Et cetera
MySQL	My Structured Query Language
SDLC	System Development Lifecycle
ACM	Association for Computing Machinery
URL	Uniform Resource Locator
UAT	User Acceptance Testing
ER diagram	Entity Relationship Diagram

Table: List of abbreviations.

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1. Introduction

The Pet Buy, Sell, and Adopt Website project aims to address the growing need for an online platform that facilitates the process of buying, selling, and adopting pets. The project recognizes the limitations and inefficiencies of traditional methods of pet acquisition and aims to create a centralized digital platform that connects pet enthusiasts, shelters, and responsible breeders. The Pet Buy and Sell website is a place of sale, grooming, daycare, and purchase of pet supplies. The website can be positive because it can foster a sense of spirit to keep pets. The Pet Buy, sell, and Adopt website aims to make it easier for Pet Buy and Sell customers and owners to interact where pet Shopping is available without time limits and locations.

1.1. Project Background

With more individuals adopting pets as their companions, the pet sector has seen significant growth throughout the years. As a result, there is an increasing need for pet buy, sell, and adoption websites to make it easier to identify and adopt the ideal pet (Feng 2022). Without having to physically visit a pet shop or shelter, these websites provide pet owners with a convenient platform to buy, sell, or adopt pets. Furthermore, the website will allow pet owners to discover the ideal pet to adopt, making pet ownership easier and more accessible. The project will be implemented successfully and achieve its goals of offering a platform for pet lovers to purchase, sell, and adopt pets in a safe, convenient, and entertaining way by considering these topics and applying their pertinent concepts.

The target audience for a pet buy, sell, and adopt website would include:

- Pet owners: Individuals who are looking for a marketplace to sell or adopt their pets.
- Prospective pet owners: Individuals looking for a platform to purchase or adopt a pet yet are interested in doing so.
- Pet Organizations: Organizations that provide pets for adoption and are searching for a platform to connect with potential adopters include animal shelters and rescue groups.
- Pet breeders: Individuals who raise animals for sale and are looking for a venue to promote and market their animals.
- Pet-related companies: They may desire to advertise their goods or services on the website. Examples include pet shops, vets, pet food manufacturers, and pet grooming services.

- Animal Lovers: People who are interested in animals and wish to keep up with the most recent developments in the animal world.

Pet owners, potential pet owners, animal shelters and rescue groups, pet breeders, pet-related businesses, and animal lovers who are interested in buying, selling, or adopting pets or learning more about pet-related topics would be the target audience for a pet buy, sell, and adopt website.

1.2. Project Aim and Objectives

The project aims to create a centralized platform connecting pet owners, buyers, and individuals looking to adopt pets. The aims are listed below:

- Develop a user-friendly website for purchasing, selling, and adopting pets.
- Key Features:

1. Search Bar: Implement a search function to allow users to find specific pets based on their preferences.
2. Purchase and Selling: Enable users to buy or sell pets through secure and straightforward transaction procedures.
3. Pet Variety: Cater to all types of pets, including dogs, cats, birds, fish, and small mammals, to accommodate a wide range of pet enthusiasts.

- Prioritize a user-friendly experience by ensuring intuitive navigation, clear information display, and an appealing interface.
- Offer a valuable tool for pet owners seeking suitable pets and individuals looking to find loving homes for their pets.
- Implement a commission-based structure where a small percentage of the transaction amount serves as the website's revenue.
- Simplify the process of pet transactions and provide a convenient and reliable platform for pet enthusiasts.
- Facilitate responsible pet ownership, connect pet lovers, and contribute to the well-being and happiness of pets and their owners.

To accomplish this goal, I'll take the following actions:

- Carry out research: To identify the target market, the most popular pet breeds and species, and the most effective business models, conduct research on the pet industry, including current websites, competitors, and market trends.
- Create a strategy: Create a detailed strategy including all the components of the website, such as user registration, search and filter options, pet listings, payment methods, and customer care.
- Create the website: Create the website utilizing contemporary web development best practices and technologies, making sure it is user-friendly, responsive, and practical.
- Test: Test the website carefully to make sure that it performs as planned and that all its features and functionalities are operational.

The project's overall goal is to build a profitable pet purchase, sell, and adoption website that helps animals, their owners, and the pet business while also offering people a useful service.

1.3. Project Development Methodology

The strategy that will be used in software development is Agile (Scrum) in this part, which focuses on the software life cycle. A Gantt chart has been created using the project's methodology, listing the activities in chronological order (please go through Appendix 1).

1.3.1. Software Requirements Engineering and Solution Specification

The engineering elicitation methods to be used in the project are background research and interviews. The alternative methods that were not used are listed below along with their justifications:

Technique chosen:	Advantages	Disadvantages	Justification
Client Interviews	• With the relevant stakeholders, it is possible to swiftly identify a broad perspective of the problem domain and	• may call for the opinion of management, who will not be around too much. • can be pricey. • It is crucial to have a	It is critical to take the client's input into account when creating the product's specifications. This method explains the justification for the requirement for a solution system.

	receive immediate input. • The interviewee's genuine feelings regarding a project are frequently conveyed through nonverbal cues. • Although preparation is crucial, the subjects are interchangeable.	comfortable interview approach that makes the client feel at ease.	This common method of elicitation enables the discussion of all pertinent subjects.
Background Research	• compares the client's present system against comparable software that is on the market. • Comparison can identify features that should be included in the suggested solution. • ensures that the problem domain is well-known. • Finding a low-cost result is quick and simple.	• The data might not be current. • The information's quality could be poor or irrelevant.	For client meetings, it is crucial to prepare as much as you can. To accomplish this, background research is a quick and affordable option. An excellent method to stay competitive is to research comparable software systems already available on the market.

Table 1.3.1 - 1: Chosen Software Requirements Engineering and solution specification methodologies.

1.3.2. System Analysis and Design

The creation of a website depends heavily on system analysis and design. It entails a methodical approach to comprehending user needs, determining system functionalities, and building an effective system that is also user-friendly. As the system analysis and design methodologies to be used in the project, Event Analysis, Commands, Queries and Constraints, Wireframes, and Screen Mock-ups were selected. The alternate methods utilized, together with their justifications, are listed below:

Technique chosen:	Advantages	Disadvantages	Justification
Significant Event Analysis	• Each event includes information about the classes being used and the properties that go with them. • Emphasize the classes that alter the status of the system. • Emphasize the redundant attributes that should be reduced.	• concentrates on the current system.	This method examines the routines that will occur and lists the classes required to make them possible.
Commands, Queries, Constraints	• Describes each class and its associated procedures, properties, and logical principles. • intended to explain how	• A new formatted presentation of the same information as SEA is required.	compliments TA and SEA methods while going into further detail.

	class limitations specified by the requirements definition might be permuted.		
ER Diagram	• A simple and understandable visual representation of the database structure is offered by ER diagrams. • An effective ER diagram can result in quick database queries.	• While ER diagrams are often straightforward to understand, they can get complicated for very big and sophisticated databases. • A high-level perspective of the data model is provided via ER diagrams.	A relational database's structure can be conceptualised, designed, and communicated using ER diagrams. They provide database development with clarity, simplicity, and a single language.
Wireframes	• Traditionally, to take layout, behaviour, and navigation into account, the customer and the developer write paper-based designs. • gives the eventual product's design and implementation a head start. • Ideally, it won't take too long.	• increases the documentation's layering. • Later, it may be difficult to read design notes.	Using in selecting the system's appearance and feel, assuring excellent customer satisfaction.

Screen Mock-ups	• The system narrates the purpose, error messages, necessary inputs, and other pertinent information for each screen. • Communication with clients is crucial. • Important design elements are precisely defined for implementation, thus saving time. • To guarantee a successful product, features of heuristic evaluation should be considered.	• can take a lot of time to do effectively. • A client may still request modifications.	These design pages may take some time, but the information they offer will be invaluable for implementing the design. The papers can be consulted if there are any questions about functionality.

Table 1.3.2 - 1: Chosen system analysis and design methodologies.

1.3.3. System Implementation

The developer has determined that PHP is the ideal language for the system based on his research into several coding languages. This is because the Laravel framework has so many tools at its disposal. Because PHP in Visual Studio is GUI-based, system layouts may be created quickly, freeing up time for the functionality of the system.

1.3.3 System Implementation			
Technique chosen	Advantages	Disadvantages	Justification

User Interface	• leverages the fundamentals of user interaction with computers. • A need for non-technical consumers.	• Favourite designs may change to reflect the times. • Every conceivable navigational need must be considered by the developer.	The system's user interface has a significant impact on the project's success or failure. Users must be motivated to keep using the system.
Programming	• making sure the system can perform the tasks for which it was created. • possesses the ability to produce extraordinarily sophisticated software.	• process that is prolonged. • Expensive. • requires a very solid understanding of object-oriented software.	This is essential to ensure that all required functionality is present in customized software.
Code Comments/Documentation	• aids in future system development and maintenance. • particularly useful when numerous developers are collaborating on the same project. • defines a code in further detail. • divided into important categories.	• can make programs significantly larger and text appear thick.	Code comments are helpful while working on a project for a lengthy period. Sometimes the intent behind a piece of code is not immediately obvious. Working on various projects will inevitably lead to conflicts, but documentation will make them easier to resolve.

Table 1.3.3 - 1: Chosen System Implementation Methodologies.

1.3.4. Database Development

The creation of a relational database for the system would be the best use of time. The PHP program is compatible with relational databases, which are quick. If necessary, the database can be readily changed.

An open-source relational database management system is called MySQL (RDBMS). One of the most utilised and admired database systems worldwide. Because it is available on numerous platforms and is renowned for its dependability, performance, and usability, MySQL is a popular option for many web applications and software projects.

1.3.4 Database Development			
Technique Chosen	Advantages	Disadvantages	Justification
Relational Database	• Flexible and simple to extend. • good for storing a lot of data in alphanumeric format. • Queries make it simple to obtain data, and tables can be connected to integrate data.	• Many-to-many relationships should not be used; hence link entities are needed. • The same data does not equate to the same object. • difficulties dealing with complex data types, like multimedia	For this project, a relational database is utilised, and MySQL is the language of choice.

Table 1.3.4 - 1: Chosen Database Management System.

1.3.5. System Testing

Unit Testing, Blackbox Testing and Acceptance Testing were chosen for the system testing techniques to be implemented in the project.

1.3.5 System Testing			
Technique Chosen	Advantages	Disadvantages	Justification
Unit Testing	• "Functional decomposition" is the focus.	• It is challenging to test smaller portions of the programme on their own. • involves writing a lot of extra code.	The system is broken down into working "units" and put to the test.
Black Box Testing	• The tester does not require technical expertise in the operating procedures. • checks the degree to which the needs of the software have been met. • reveals contradictions in the specification.	• Consuming, especially if testing for every potential outcome. • cannot test code sections. • likely to occur with numerous users in repetition.	highlights any problems that occur during the process development stage, starting with the specification.
Acceptance Testing	• assures client fulfilment and recognises it as a success.	• If the end solution doesn't resolve the issue, the client could not consider the project a success.	Client happiness is the highest honour for a successful software solution.

Table 1.3.5 - 1: Chosen System Testing.

1.3.6. System Evaluation

Interview/testing feedback should be documented to evaluate the usability of the software with details of how much training was needed for users to feel proficient with the system.

1.3.6 System Evaluation		
Technique Chosen	Features	Justification
Usability Evaluation	• The customer and other stakeholders are asked to test the finished software product and provide feedback on the project's success. • Describes the degree to which the non-functional requirements were put into practise.	This will detail the non-functional requirements area's successes and failures.
Project Review	• A determination of the software solution's success is made after considering the testing findings, usability input, and its ability to accomplish the original project goals and objectives. • Details how much of the system's goals were accomplished. • includes system improvements going forward.	Any piece of writing should have both strengths and shortcomings identified. This method records the final product's results.

Table 1.3.6 - 1: Chosen System Evaluation Methods.

2. Literature Review

Introduction:

The following survey of the literature offers an overview of the scholarly work and current research on practices for buying, selling, and adopting pets. To shed light on the various facets of online pet marketplaces and adoption platforms, the review seeks to investigate major themes and findings in the field.

1. Adoption Process and Consumer Behaviour:

Understanding the adoption process and customer behaviour in the context of pet buy, sell, and adopt websites has been the subject of several research. (Sazkaya 2018) research highlighted the significance of ease, diversity, and emotional resonance in influencing consumer interest and adoption decisions. They discovered that internet platforms enhance adoption rates by giving

potential adopters a variety of options and enabling them to bond with animals based on personal preferences.

2. Trust and Security:

Pet buy, sell, and adopt websites are successful in large part because of qualities like trust and security (Brzezinski et al. 2021). Studies have looked at the indicators and data that support the development of buyer-seller trust. According to (Nwaiwu et al. 2020), the availability of reviews and ratings, accurate pet descriptions, and verified seller profiles all help to build customer trust. To reduce potential dangers and concerns, they also emphasized the importance of secure communication channels and transaction processes (van den Burg et al. 2022).

3. User Experience and Interface Design:

Websites for buying, selling, and adopting pets depend heavily on user experience (UX) and interface design. The importance of a user-friendly interface, clear navigation, and responsive design for boosting the overall user experience was underlined in research by (Orlando Putra and Setiawan 2019). To enable simple browsing and fruitful pet adoptions, they underlined the need for precise categorization, search filters, and smooth communication channels.

4. Ethical Considerations and Animal Welfare:

The ethical issues and animal welfare aspects of buying, selling, and adopting pets have been the subject of several studies. The need for responsible pet ownership and the importance of internet platforms in encouraging moral adoption practices were stressed in research by (Philpotts et al. 2019). They did mention the value of partnering with reputable shelters, advocating spaying, and neutering, and offering instructional materials to protect the welfare of animals and stop unethical behaviour.

5. Collaborations and Partnerships:

The literature has begun to pay more attention to collaborations and partnerships between online platforms and shelters and rescue groups. Successful collaborations between online pet marketplaces and animal welfare organizations were examined in research by (Horecka and Neal 2022). They highlighted critical components that are necessary for successful partnerships that

enhance adoption outcomes and advance animal care, including open communication, streamlined processes, and pooled resources.

Conclusion:

The literature study emphasizes the multifaceted character of pet adoption, purchase, and sale procedures. It emphasizes how crucial user experience, convenience, trust, and ethical considerations are to the success of online platforms. Stakeholders may establish platforms that satisfy the demands of both adopters and sellers while encouraging responsible pet ownership and animal welfare by comprehending and incorporating these results into the creation of pet buy, sell, and adopt websites. Future studies should keep looking into new patterns and assess how well treatments work in this changing field.

3. Requirements Engineering

Requirement Engineering is the process that investigates and clarifies problem areas and needs before creating designs and documenting the system attributes that will satisfy those requirements. The sub-sections that discuss interview tactics, interview results, problem domain descriptions, reviews of related software systems, and the design of relevant rules are all examples of elicitation activities. It also discusses design restrictions, commercial constraints, performance needs, and functional requirements.

3.1. Elicitation Activities

3.1.1. Interview plans

Client Interview 1

a. User Experience:

- What led you to conduct an online pet search?
- What are the main things you look at when you browse pet listings?
- How significant is it to you to have thorough pet profiles that include details about their health, and behaviour?
- What features would you want to see in a pet buy, sell, or adoption process?

b. Trust and Security:

- How crucial is it to you that you feel safe when interacting with pet adopters or dealers online?
- What facts or evidence would you need to believe a pet seller or listing?

3.1.2. Interview findings

Stakeholder	Question number	Questions and Client Responses
Mr. Khem Raj Sharma (Buyer):	01	What led you to conduct an online pet search? - He commented on how convenient it was to look for pets while relaxing in his own home. Also said that Internet pet stores and shelters typically have a larger selection. Mentioned the availability of viewing many postings, comparing possibilities, and researching various breeds at his own pace.
	02	What are the main things you look at when you browse pet listings? - When looking through pet listings, the participant mentioned several crucial elements. These included precise and thorough descriptions of the pets, along with details on their breed, age, health, and temperament. For visual evaluation, high-quality images or videos of the animals were also mentioned as crucial. A participant also emphasized the significance of being fully informed about the adoption procedure and the adoption costs in advance.

	03	How important do you think it is to maintain detailed pet profiles with information on their behaviour and health? - The participant stressed how crucial it is to feel confident while interacting with pet adopters or vendors online. They stated that they wanted verified, reliable profiles with contact information and background information about the sellers or adopters. A participant also emphasized the necessity of safe payment methods and secure communication routes. Reviews and ratings from prior adopters were regarded as trust markers that boosted confidence in the deal.
	04	What features would you want to see in a pet buy, sell, or adoption process? - The participant listed a few crucial search parameters and filters when seeking a pet. Location, species, breed, age, size, and gender were among them. Participants also expressed interest in filters that might be applied depending on characteristics like energy level, compatibility with kids or other pets, and particular requirements based on one's living circumstances or lifestyle. He stressed the value of having a selection of filters available as well as the simplicity with which they could simply personalize and narrow their search.

	05	- How important is it to you that you feel secure while communicating with pet dealers or potential adopters online? - The participant stressed the significance of feeling safe when interacting with pet adopters or dealers online. He said that crucial aspects of the decision-making process were security and trust. The necessity for procedures like verified seller profiles, background checks, and transparency regarding the vendor's reputation or record was mentioned by a participant. Additionally, it was thought that building confidence required secure communication channels and financial transaction safeguards.
	06	What facts or evidence would you need to believe a pet seller or listing? - Participants mentioned several measures and information that would contribute to their trust in a pet listing or seller. These included clear and accurate descriptions of the pet's history, behaviour, health records, and any known issues or special needs. Participants also highlighted the importance of having access to photos or videos that genuinely represent the pet's current condition. Verified seller information, testimonials or reviews from previous adopters, and the ability to ask questions or communicate directly with

		the seller were seen as trust-building elements.
	07	- How do you want to communicate with potential customers or adopters during the process? - The participant stated that they would prefer to interact with sellers or adopters using a variety of communication methods. He suggested that options like phone conversations, emails, or direct messaging would be practical and enable more individualized encounters. The necessity of prompt and attentive communication from sellers or adopters was also stressed by the participants since it inspires confidence and enables them to obtain all the information they need before deciding.
	08	What parameters or search filters are crucial to you while looking for a pet? - The participant mentioned a few crucial search parameters or filters when seeking a pet. Location, breed, age, size, gender, and compatibility with other pets or kids were among these factors. There were also additional filters like specialized training requirements, activity intensity, or coat type. Participants emphasized the value of having a variety of filters to customize the search to their needs and interests.

	09	- How important do you believe it is to have a system in place to improve or evaluate pet stores or shelters? - A participant stated that they would be interested in being able to rate or review pet stores or shelters. He thought that evaluations and ratings contributed to transparency and accountability by offering insightful observations from prior adopters. A participant said that having access to other people's experiences and the ability to evaluate sellers' or shelters' reputations would have a significant impact on their decision-making process and foster more trust in the platform.
	10	Do you have any recommendations for enhancing these websites' user interfaces or navigation? - Participants offered ideas to enhance pet buying, selling, and adoption websites' user interfaces and navigation. One suggestion was to have a simple, user-friendly interface with prominently displayed menu selections and search capabilities. He also proposed adding visual cues, such as symbols, or labels, to help consumers grasp the features and functionality of the website. For a seamless user experience, a participant raised the significance of responsive design that works well with various devices and screen sizes.

Table 3.1.2 - 1: Interview table.

- How would you prefer to interact with potential buyers or adopters during the process?

c. Website Functionality:

- What parameters or search filters are crucial to you while looking for a pet?
- How vital do you think it is to have a way to enhance or review pet shops or shelters?
- Do you have any recommendations for enhancing these websites' user interfaces or navigation?

I asked participants for their permission to participate and record courteously while maintaining their confidence. The interviews will offer insightful information that will help shape the creation of the pet buy, sale, and adoption website and guarantee that it meets the demands of both users and animal shelters.

3.1.3. Other problem domain research

Research on existing systems as well as extra sources, such as pertinent legislation that will help comprehend the problem domain is included in this topic.

3.1.3.1. Comparable Software System Review

Existing online resources like Petfinder, Adopt-a-Pet.com, Pet Store Nepal and Rescue Me! make it easy to adopt a pet. Additionally, a great deal of study has been done about online pet adoptions and how these platforms affect adoption rates and animal welfare. In my prior response regarding research papers on this subject, I cited a few of these studies.

No one solution has been widely adopted or regarded as perfect for the ongoing issue of pet buy, sell, and adopt websites. However, a lot of online resources have been effective in promoting pet adoptions and raising adoption rates. Research studies have also been carried out to assess the efficiency of these platforms and pinpoint areas for development. The current platforms have

significantly improved access to pet adoptions and promoted animal welfare, even though there is still potential for improvement.

A conventional set of web development tools, databases, and third-party APIs would serve as the website's underlying technology for buying, selling, and adopting pets. Web development tools like HTML, CSS, and PHP would be used to create the website. To keep track of data about available dogs, prospective adopters, and previous adoptions, a database would be required. Additional features like payment processing or location-based search functionality may be integrated using third-party APIs. Additionally, the website may leverage machine learning or artificial intelligence (AI) technology to enhance the adoption process or give users tailored recommendations. The page that inspired me to build a better and more challenging website is (Mount Everest Kennel Club 2023).

<u>A brief review of Pet Store Nepal:</u>

Homepage

Visitors to PetStoreNepal.com are welcomed by a neat and eye-catching site. The branding and logo of the website are displayed at the top, lending credibility. The design is user-friendly and has a simple navigation menu with sections for "Dogs," "Cats," "Fish," and "Small Mammals."

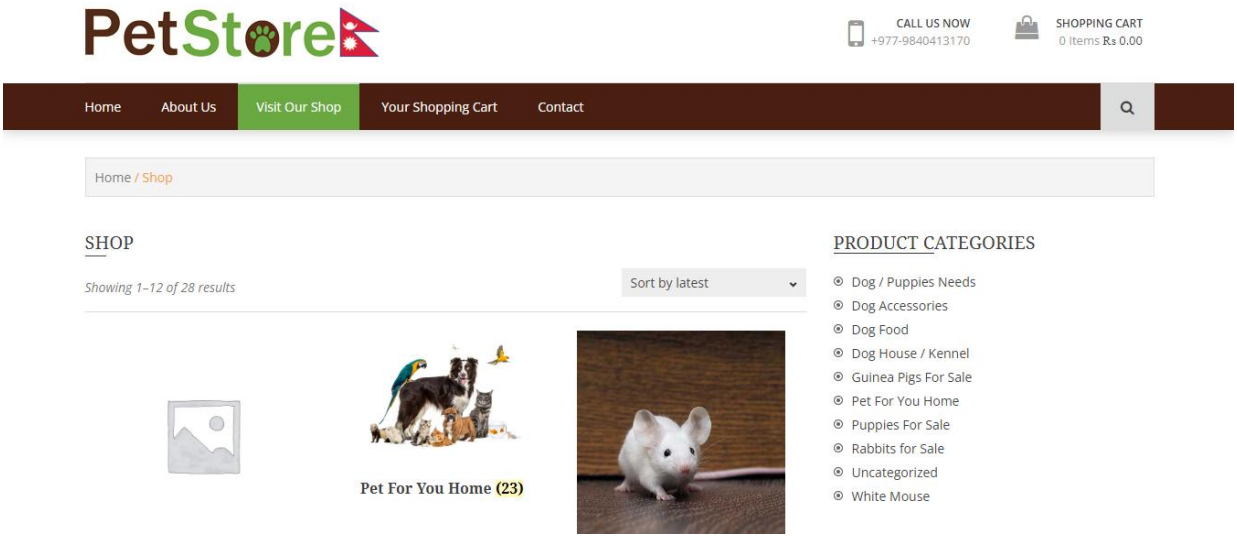

Figure 3.1.3.1 - 1: Home page of Pet Store Nepal (petstorenepal 2023).

Product Listings

Users looking for specific pets or pet-related products can easily navigate to their desired category. The product listings are well-organized, displaying high-quality images, product names, prices, and brief descriptions. The use of filters allows users to narrow down their search based on criteria such as price range, pet type, and location.

Product Details

When you click on a product, a comprehensive product page opens. Users can access in-depth facts on the pet or product, such as specifications, seller information, location, and availability, here. Users are encouraged to make purchases with the prominently displayed "Add to Cart" and "Buy Now" buttons.

User Registration and Login

Users must register or log in to make buying and selling possible. The registration process is simple to use and only asks for the necessary information. For convenience, users can sign in using their social network accounts.

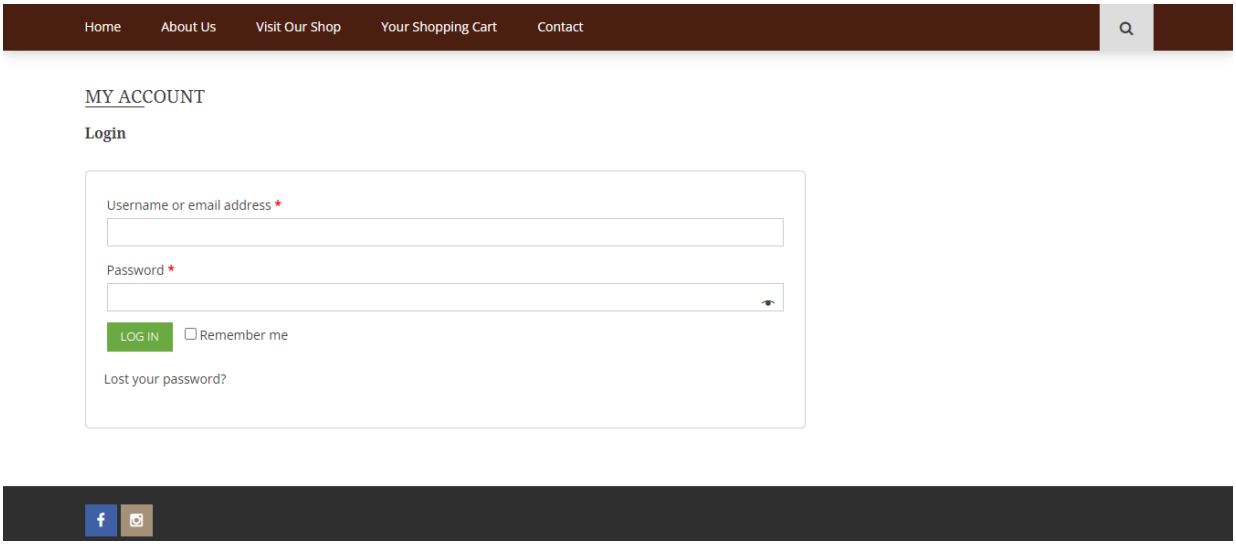

Figure 3.1.3.1 - 2: Login Page of Pet Store Nepal.

Cart and Checkout

The shopping cart is easily accessible, displaying a summary of items, prices, and an option to apply discount codes. The checkout process is streamlined, with clear steps for shipping and

payment information. Multiple payment options, including credit/debit cards and online wallets, are available to cater to user preferences.

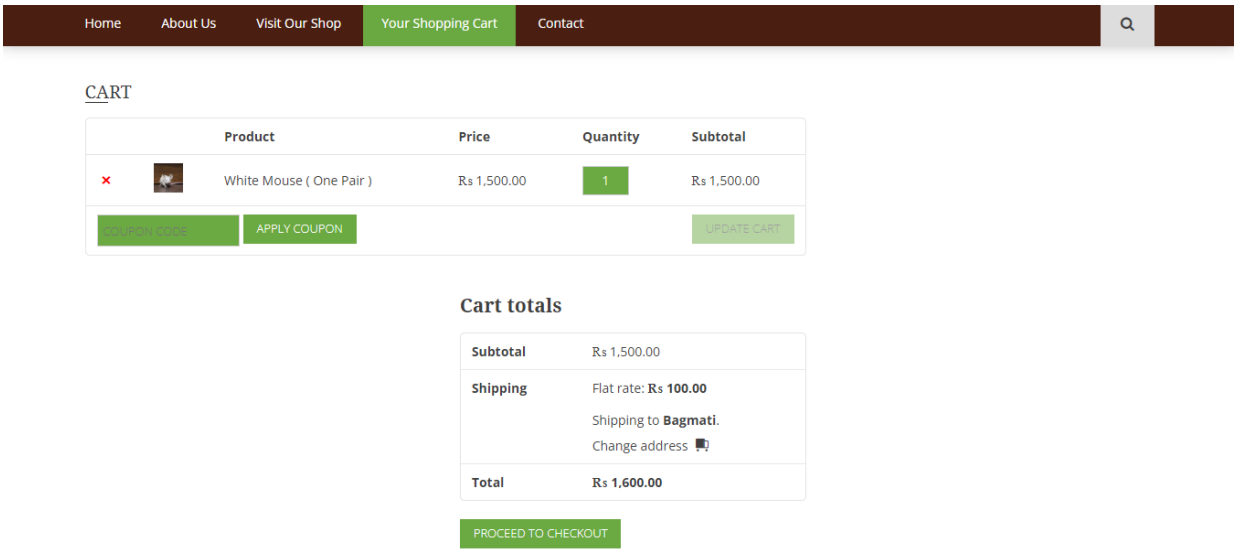

Figure 3.1.3.1 - 3: Cart of Pet Store Nepal.

Security

PetStoreNepal.com places a high priority on user data security and payment protection, enforcing SSL encryption and other security protocols to secure user information.

Conclusion

PetStoreNepal.com provides an easy-to-use and aesthetically pleasing platform for purchasing and selling animals and animal-related goods. Its user-friendly layout, thorough product listings, and safe payment methods all enhance the shopping experience. The website effectively meets customers' demands, whether they are looking for a new pet companion or pet supplies.

3.2. Requirements Specification

3.2.1. Problem Domain Description

Professional, social, economic, and legal issues

The following social, legal, and economic aspects of the pet buy, sell, and adopt website should be considered:

Social Implications:

- Promoting animal welfare: By offering a forum for responsible pet adoption, purchasing, and selling, the website can contribute to the promotion of animal welfare.
- Increasing pet ownership: The website might persuade more individuals to adopt pets, which can be advantageous for their social and physical well-being.
- Reducing animal homelessness: By offering a platform for pet adoption, the website can aid in reducing the number of homeless animals.

Legal Implications:

- Faithfulness to applicable animal welfare laws: To guarantee that pets are treated morally and humanely, the website must abide by all applicable animal welfare laws and regulations.
- Compliance with privacy laws: The website must also abide by privacy laws and regulations to protect user data.
- Liability for pet transactions: To safeguard all parties involved, the website may encounter liability difficulties relating to pet transactions.

Economic Implications:

- Increasing the pet industry: By offering a platform for pet adoption, purchasing, and selling, the website can increase the pet industry.
- Revenue generation: The website may make money in several ways, including through transaction fees, sponsorships, and advertising.
- Competition firms, and current pet-related businesses: The website may encounter competition from these firms; thus, it is important to build effective marketing and business strategies to stay competitive.

Therefore, numerous social, legal, and economic ramifications of the pet buy, sell, and adopt website should be carefully considered during the building and management of the website.

I plan to adhere to numerous professional standards. I'll abide by the ethical standards for software engineering established by organizations like the Association for Computing Machinery (ACM) to guarantee that the project is created and run ethically. To make sure the website satisfies the needs and expectations of its users. I will adhere to user-centered design principles.

System Development Lifecycle (SDLC): To make sure that the website is produced in a disciplined and methodical way, I will adhere to the SDLC approach. To make sure the website is error-free and up to par, I'll put quality assurance procedures in place. To guarantee that the project is finished on schedule, under budget, and meets expectations, I will adhere to project management best practices. To safeguard user data and ensure compliance with pertinent laws and regulations, I shall put in place the necessary data privacy and security measures. To keep the website current and satisfy changing customer demands and expectations, I will take a continuous improvement approach.

I can make sure that the pet buy and sell website is created and run professionally and ethically, satisfies all standards, and offers value to its users by adhering to these professional criteria.

3.2.2. Functional Requirements

The functional requirements planned for the pet buy, sell, and adopt website are:

1. All users have access to browse the products throughout the website.
2. Any user can register on the website.
3. Any registered user can log in to the website.
4. When a user logs in with the wrong username or password it gives a flash message to enter the correct username or password.
5. Every authenticated user has a profile with a default picture.
6. All the product modules should have a search field. This search field should give results even when there is only some part of the name.
7. Authenticated users can add any desired product to the cart, remove the products from the cart, and confirm the order.
8. Only authenticated users have access to the cart, profile, and orders pages.
9. When a user manipulates the URL and changes the product id. If the product ID is not in the database, it redirects the user to the page not found error.
10. The Admin can add, edit and delete the pet and products from the database.
11. Admin can add, edit and delete the pet and product categories from the database.
12. Admin can view the orders confirmed by the user.
13. The admin can view the order details and payment details.
14. Admin can manage any order or user as admin likes.

15. Dashboard in the admin panel to show the total number of pets, products and total income.

3.2.3. Performance Requirements

- Hardware Needed to Complete the Project:
- 40 GB of hard disk
- RAM: 1 GB
- Software Requirements for the Project's Development:
- System of operation: Windows
- Web technologies include PHP, Bootstrap4, HTML, CSS, JavaScript
- Plugins include Flat icons and jQuery.
- MySQL database
- Visual Studio Code is used as the database browser for the Laravel framework and Docker.
- Browser requirements for clients: Chrome, Mozilla Firefox, etc.

4. System Analysis and Design

4.1. Preliminary Analysis Stages

4.1.1. ER diagram.

This is an Entity relationship diagram for the pet store website where there are different entities and their related attributes which shows the relationship among the whole entities.

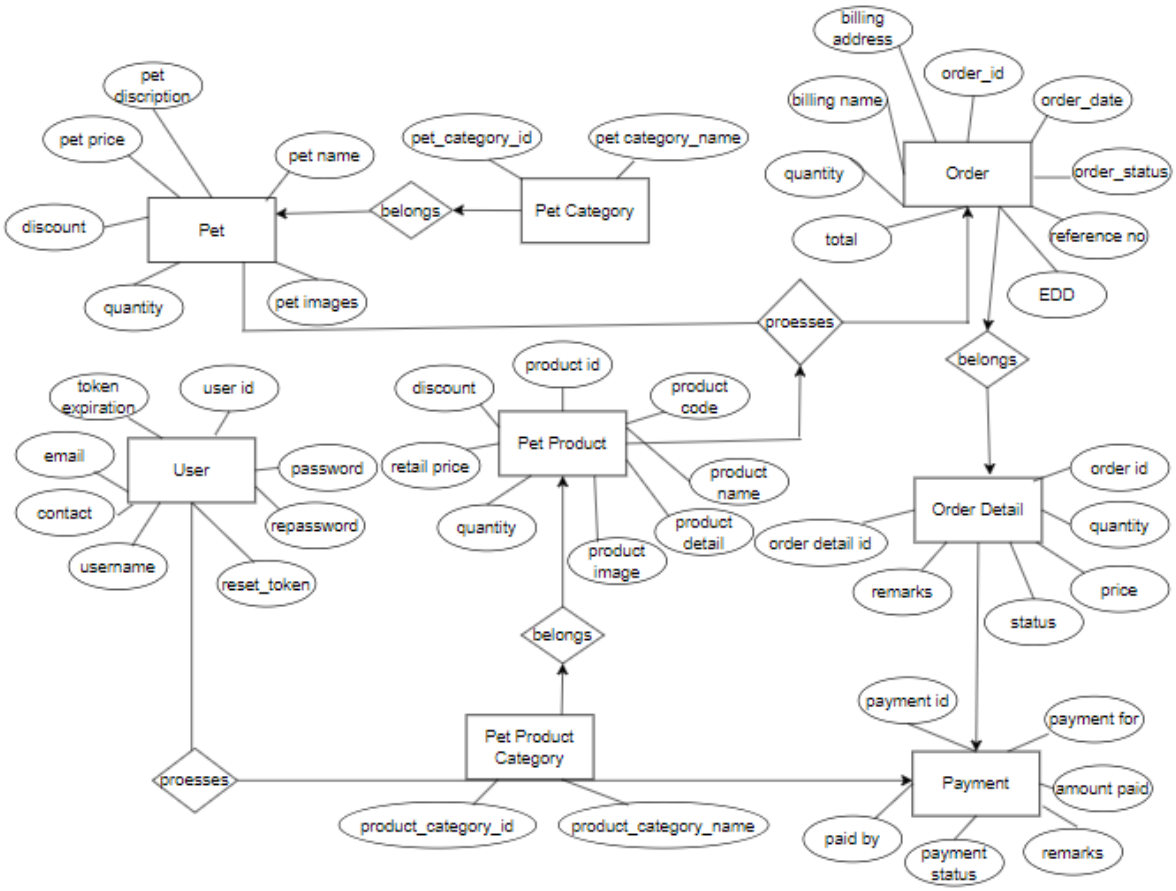

Figure 4.1.1 - 1: ER diagram of pet store.

4.1.2. Significant Event Analysis

Event	Performer	Attributes
Create pet	Admin	• Pet ID – primary key • Pet Description • Pet Category ID – foreign key • Pet images • User ID – foreign key
Create product	Admin	• Pet Product ID – primary key • Product Code • Product name

		• Product Detail • Product Category ID – foreign key • Quantity on hand • Retail Price • Discount • Status • User ID – foreign key
Search Pet	• Admin • User	• Pet id • Pet name
Search Product	• Admin • User	• Product id • Product name
Pet / product buy	• User	• Pet / Product id • Pet / Product Name • Address • Quantity • Price
Login	• Admin • User	• Email • Password
Register	• User	• Email • Username • Password • Re-password

Table 4.1.2 - 1: Significant Event Analysis of events.

4.1.3. Commands, Queries, and Constraints

Class:	Customer	**Part:** 1/1
Type of Object:	**Author:** Amir Tamang **Created:** 15-7-2023	

A user of the system, with basic permissions	
Queries:	User id, username, email, password, contact
Commands:	Display, search, add to cart, pay
Constraints:	• User id, username, contact and email cannot be null. • User id is assigned at the creation, number auto-increment and is never recycled.

Table 4.1.3 - 1: Command, queries, and constraints for Customer object.

Class:	Admin	**Part:** 1/1
Type of Object: A user of the system, with all permissions	**Author:** Amir Tamang **Created:** 15-7-2023	
Queries:	User id, username, email, password, contact	
Commands:	Display, search, add to cart, pay, add, edit, delete	
Constraints:	• User id, username, contact and email cannot be null. • User id is assigned at the creation, number auto-increment and is never recycled. • All functions are allowed by this user.	

Table 4.1.3 - 2: Command, queries, and constraints for Admin object.

Class:	Pet Category	**Part:** 1/1
Type of Object: An object that shows a pet group.	**Author:** Amir Tamang **Created:** 15-7-2023	
Queries:	Pet category id, pet category name	
Commands:	Add, delete, edit, display	
Constraints:	• pet category ID, pet category name cannot be null. • Pet category id is assigned at the creation, number auto-increment and is never recycled.	

Table 4.1.3 - 3: Command, queries, and constraints for Pet category object.

Class:	Pet	**Part:** 1/1
Type of Object: An Object representing a pet category group.	**Author:** Amir Tamang **Created:** 15-7-2023	
Queries:	Pet id, pet name, pet description, pet images, pet price	
Commands:	Add, edit, delete, display	
Constraints:	• Pet id, pet name, pet images and pet price cannot be null. • Pet id is assigned at the creation, number auto-increment and is never recycled.	

Table 4.1.3 - 4: Command, queries, and constraints for pet object.

Class:	Product Category	**Part:** 1/1
Type of Object: An object showing products.	**Author:** Amir Tamang **Created:** 15-7-2023	
Queries:	Product category id, product category name	
Commands:	Add, delete, edit, display	
Constraints:	• product category id and product category name cannot be null. • Product category id is assigned at the creation, number auto-increment and is never recycled.	

Table 4.1.3 - 5: Command, queries, and constraints for product category object.

Class:	Product	**Part:** 1/1
Type of Object: An object that represents a group of categories of products.	**Author:** Amir Tamang **Created:** 15-7-2023	

Queries:	product id, product code, product name, product detail, product image, quantity on hand, retail price, discount
Commands:	Add, delete, edit, display
Constraints:	• product id, product name, product code, product image, quantity on hand and retail price cannot be null. • Product id is assigned at the creation, number auto-increment and is never recycled.

Table 4.1.3 - 6: Command, queries, and constraints for product object.

4.2. Detailed Dynamic System Designs

4.2.1. Diagrams

4.2.1.1. Use case diagram.

This is a use case diagram for the pet store website, here the functions that normal users (customers) and Admin can do and cannot do are represented. For example: Users can register but the admin has a by default ID, which means the admin cannot register but a customer can.

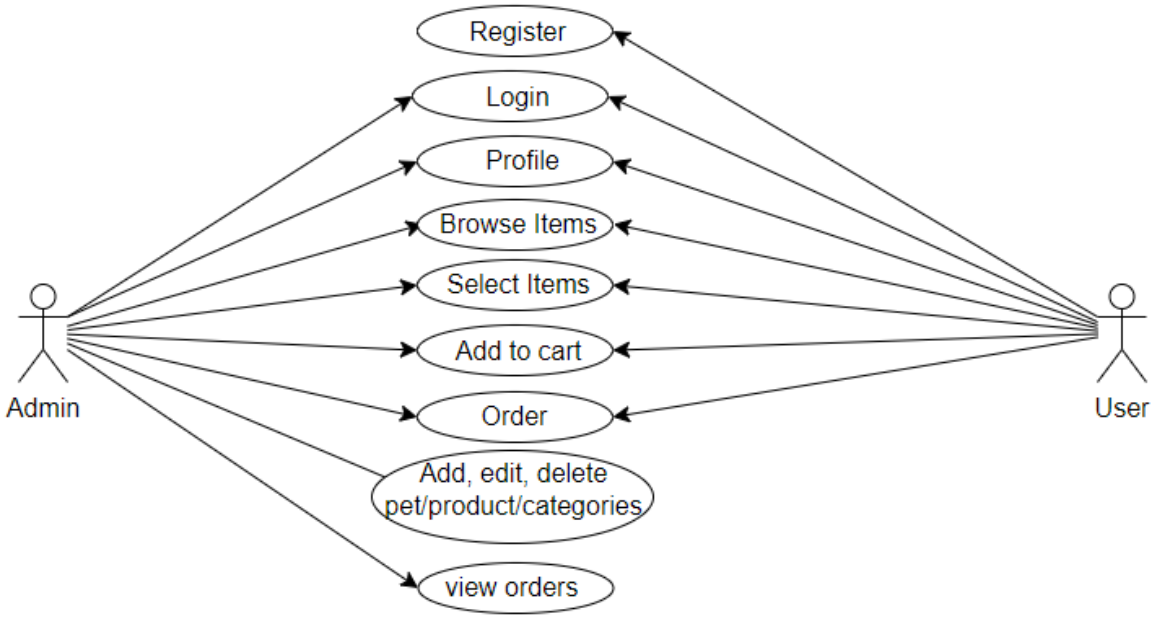

Figure 4.2.1.1 - 1: Use case diagram for the pet store.

4.2.1.2. Flow chart

This diagram represents the flow of the pet store website. The event is started and then goes to the login page. If there is no account, then the user must register and if there is an account then the user can proceed to log in which will redirect the user to their account. After that, they can browse the store, select products and add products to the cart. Before placing an order system again checks if the user is logged in or not. If not, then the user is redirected to log in, if a user is logged in then the user can place an order and proceed with payment. At last, user can continue shopping or log out according to their wish.

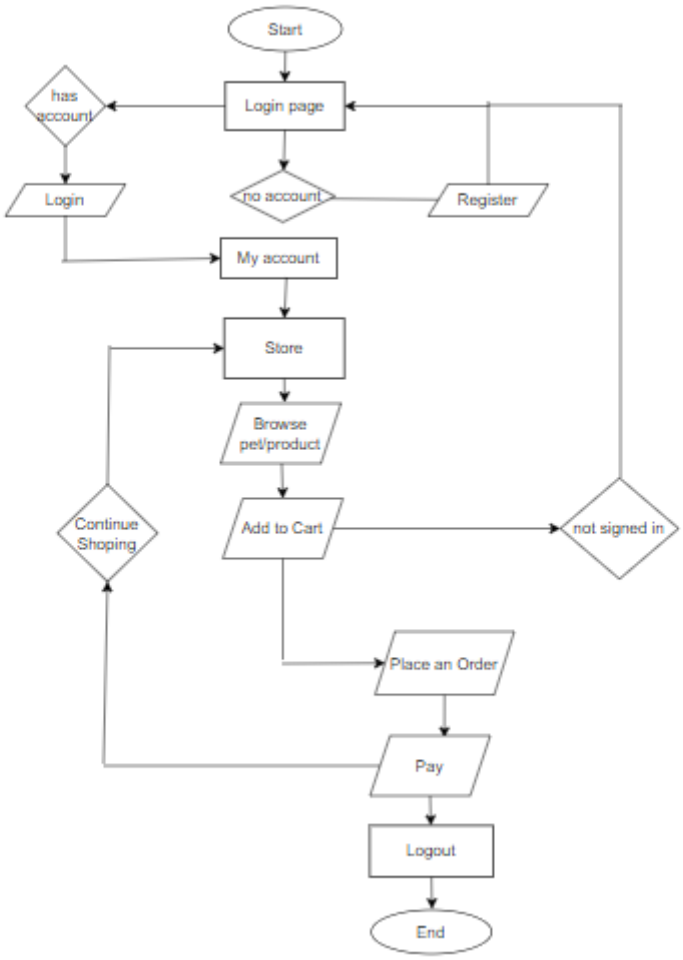

Figure 4.2.1.2 - 1: Flow Chart of the pet store.

4.2.1.3. System flow diagram

This is a system flow diagram for the admin panel which represents the functions and actions that an admin can do.

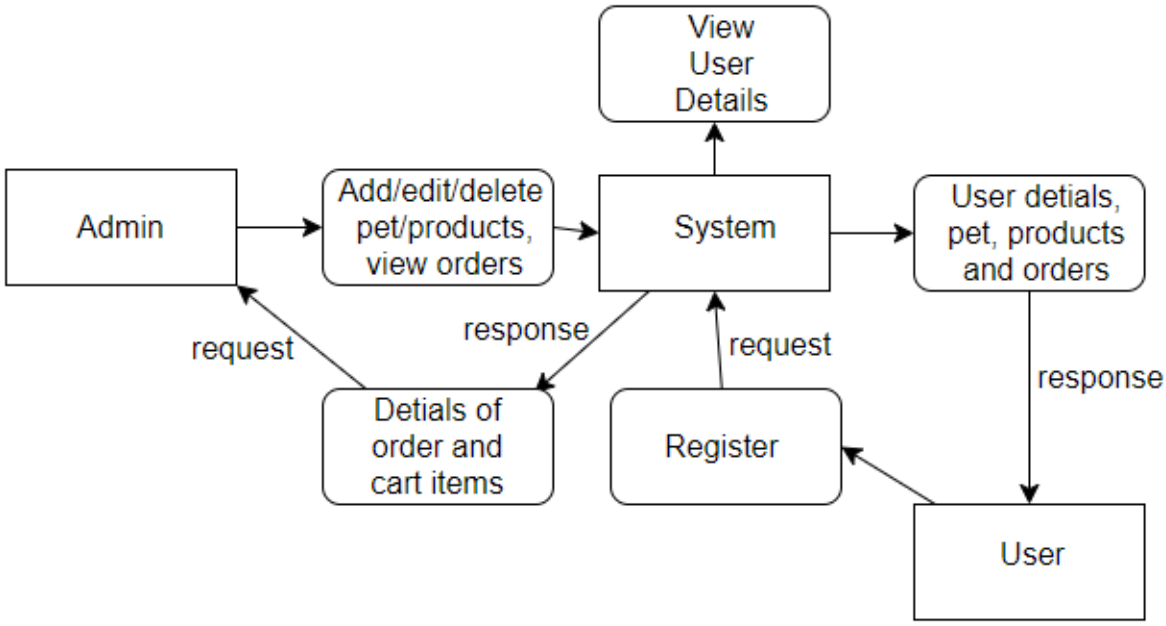

Figure 4.2.1.3 - 1: System flow diagram of the pet store.

4.2.1.4. Class diagram

This is the class diagram of the pet store which represents the class and their relationship according to their attributes.

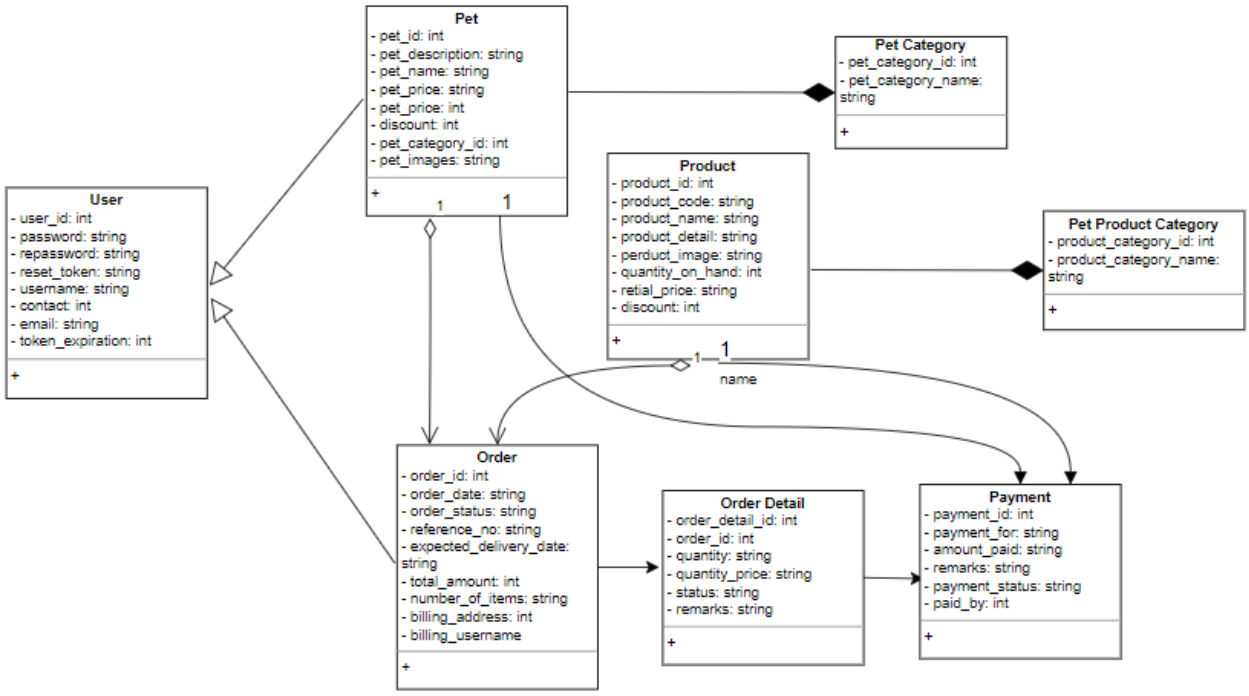

Figure 4.2.1.4 - 1: Class diagram of the pet store.

4.2.1.5. Activity diagram

This figure represents the activity diagram that a user performs on the pet store website.

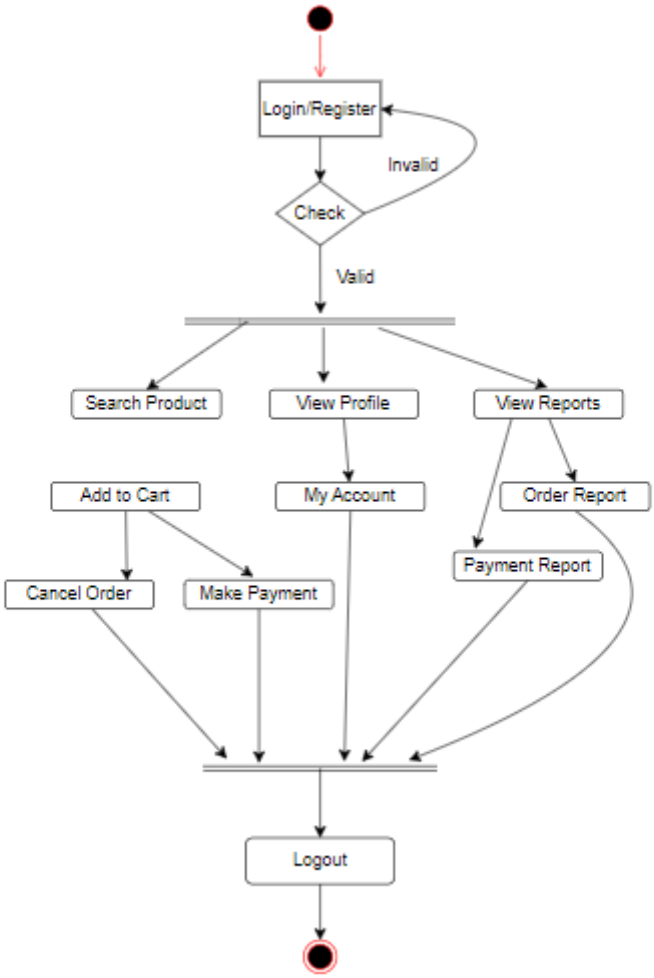

Figure 4.2.1.5 - 1: Activity diagram of the pet store.

4.2.2. Event Charts

The system's intended response to internal or external system events is shown in the charts below.

EVENTS	STORE SUPPORT	Part: 1/2
COMMENTS external occurrences that are chosen to represent representative behaviour.	INDEXING Author: Amir Tamang Create: 15-7-2023	

External (Incoming)	**Involved Object Types**
Request to create a user	Admin, Customer
Buy pet/product	Pet, product, Pet Categories, product categories, order, payment
Uploading a file	Pet, product, Pet Categories, product categories

Table 4.2.2 - 1: Event chart for External incoming events.

EVENTS	STORE SUPPORT	Part: 2/2
COMMENTS external occurrences that are chosen to represent representative behaviour.	INDEXING Author: Amir Tamang Create: 15-7-2023	
Internal (Outgoing)	**Involved Object Types**	
Logging in	Admin, Customer, user, profile	
Send an email notification to recover the password	Customer, user, profile	
Customer Downloads receipt	Pet, product, Pet Categories, product categories, price, payment, order details	

Table 4.2.2 - 2: Event chart for Internal outgoing events.

4.2.3. Object Creation Charts

Object creation is an important event in a software system. In Table 4.2.3 - 1, the classes that may create other object types are listed.

CREATION	STORE SUPPORT	Part: 1/1
COMMENTS List of classes in the systems that create objects	INDEXING Author: Amir Tamang	
Class	Creates instances of	
User	Admin, Customer	

Pet	Pet Categories
Product	Product Categories
Order	Order details, payment
Database Functionality	Read, Add, Update, Delete, Search
Store	Main Menu

Table 4.2.3 - 1: Object creation chart of Store Support

4.2.4. System Scenario Charts

SCENARIOS	STORE SUPPORT	Part: 1/1
COMMENTS a collection of illustrative examples to illustrate key sorts of system behaviour	INDEXING Author: Amir Tamang	
Create a user: Create a user and make their profile.		
Create a pet: Create a pet and assign them to their categories.		
Create a product: Create a product and assign them to their categories.		
Delete a pet: Remove a pet from their categories and delete their record		
Search a pet: Search for a particular pet using the search bar.		
User Login: User login is done by the customers and admin		
Order: Pet is listed when the order is placed.		

Table 4.2.4 - 1: System Scenario Charts for system.

4.2.5. Dynamic Diagrams

This dynamic diagram demonstrates how objects communicate with one another when the system is active. Active objects are those that send messages, while passive objects are those that only receive them. The running order is indicated by the dashed arrows.

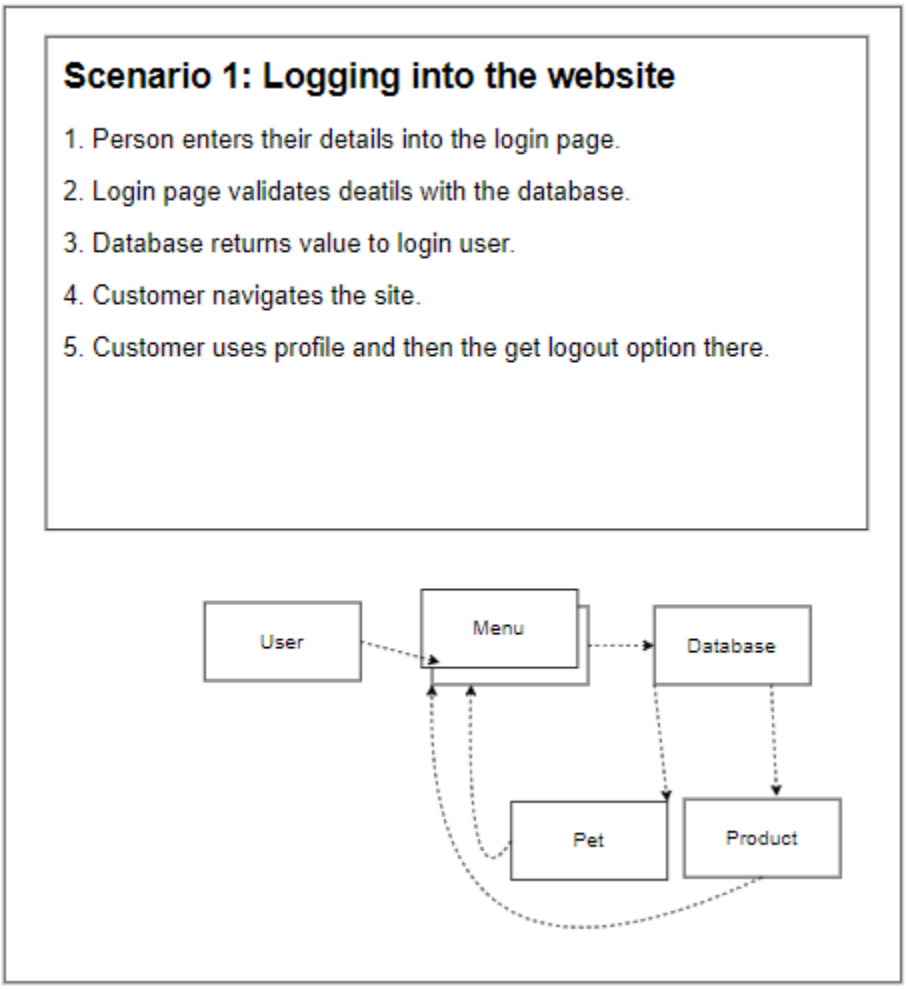

Figure 4.2.5 - 1: Dynamic Diagram – Logging in.

4.2.6. Database Design

Figure 4.2.6 - 1: Database Design for pet store website.

Figure 4.2.6 - 1 illustrates how the data will be kept at the system's back end. The figure was inspired by (inettutor 2020). Information is connected via a relational database, which frequently requires access to several tables while doing database queries. IDs are used in the design to link information together, minimising the requirement for duplicate data. The following two fields slightly deviate from normalisation standards:

- To allow querying of the relationship between the tables, the user ID column in the user table functions as the parent element to the customer and admin.
- The pet and product tables are linked to the corresponding categories tables, sharing unique identifiers to speed up data queries and avoid data duplication, both of which are features of the database system.

5. System Interface Designs

5.1. Draft Interface Designs

5.1.1. Wireframe

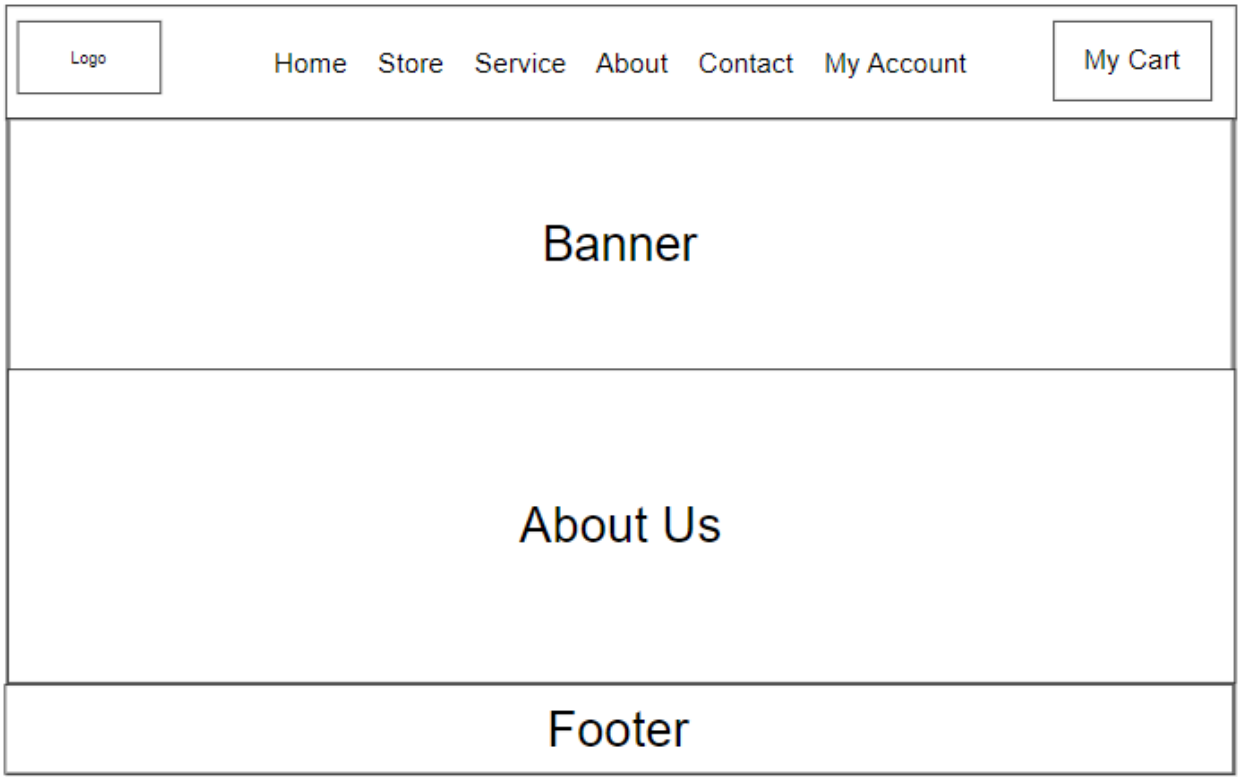

Figure 5.1.1 - 1: Wireframe of Home page.

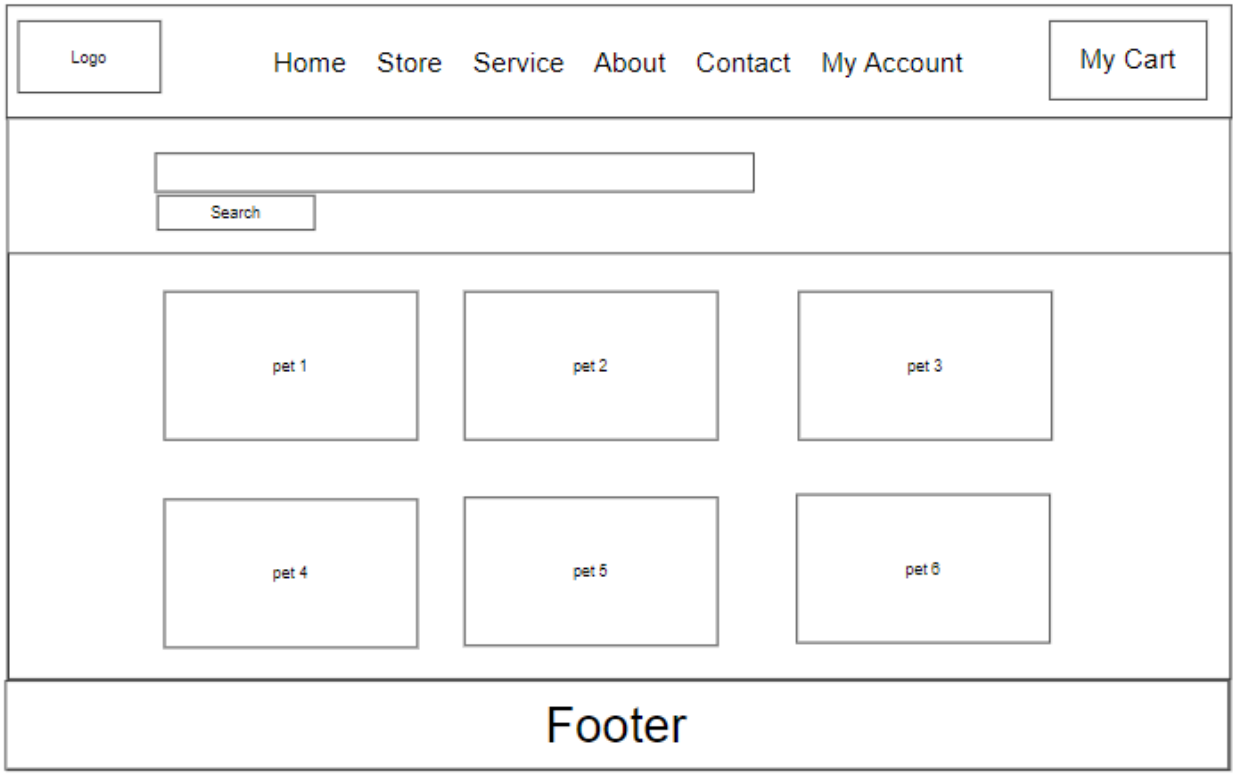

Figure 5.1.1 - 2: Wireframe of store (Draw.io 2023).

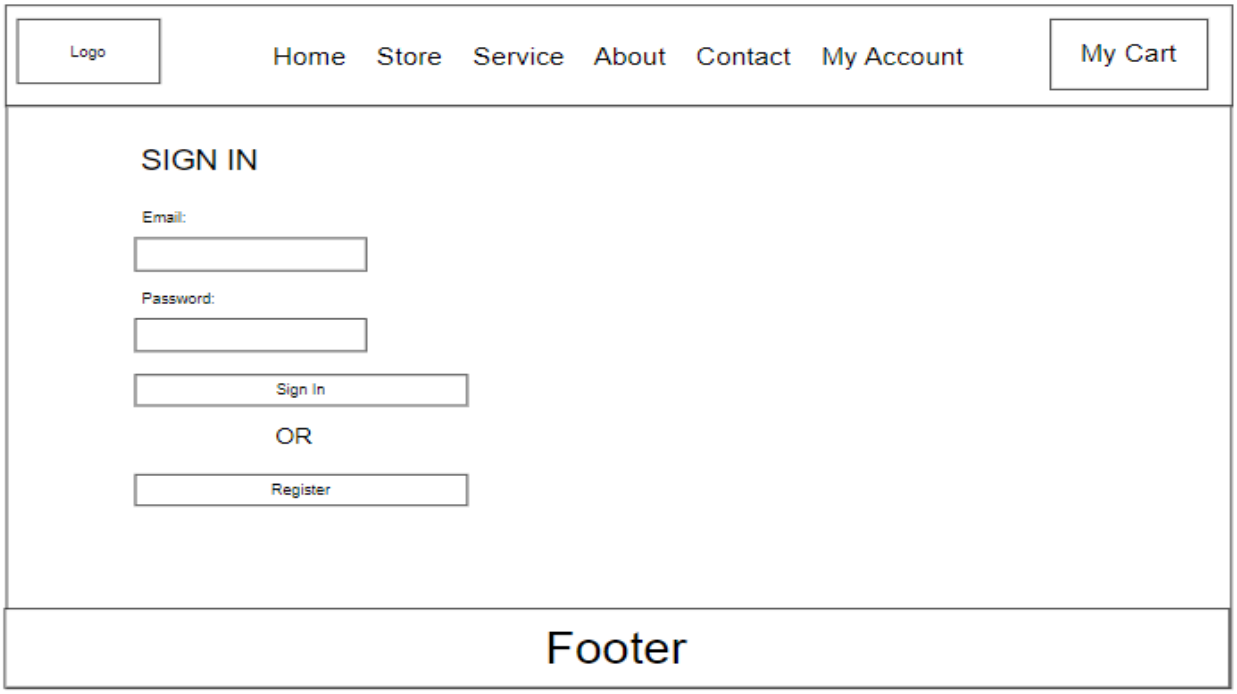

Figure 5.1.1 - 3: Wireframe of the sign-in page.

Figure 5.1.1 - 4: Wireframe of Admin dashboard.

5.1.2. System Screen Mock-ups

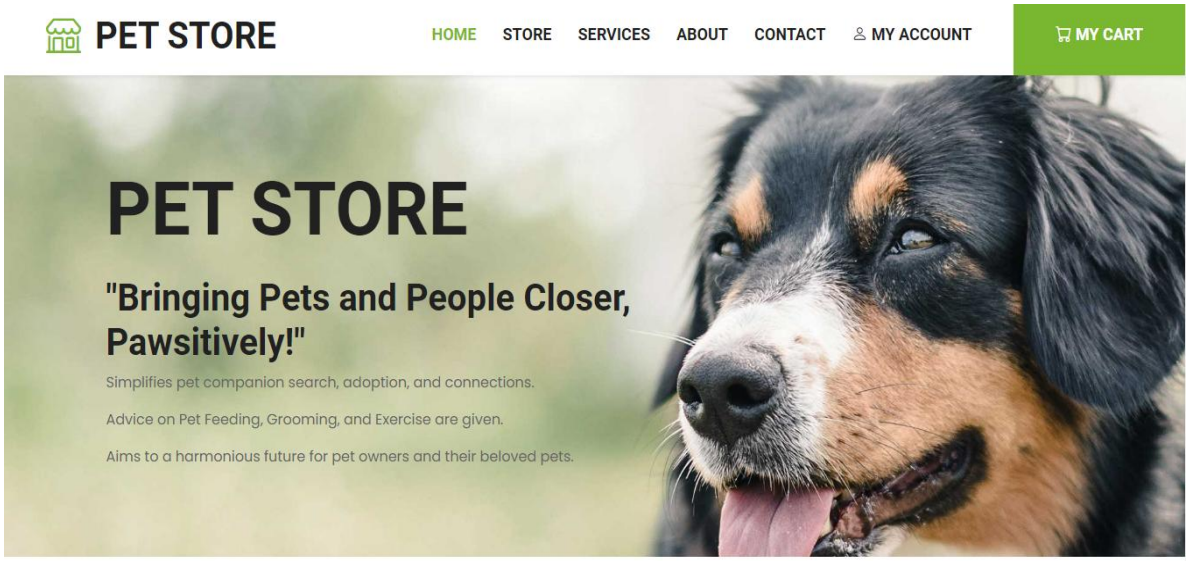

Figure 5.1.2 - 1: Home Screen Mock-up (HTML Codex 2022).

This is well well-designed form of the home page of the pet store website. From here user can navigate to different pages such as Home, Store, Services, About, Contact, My Account and My Cart at last.

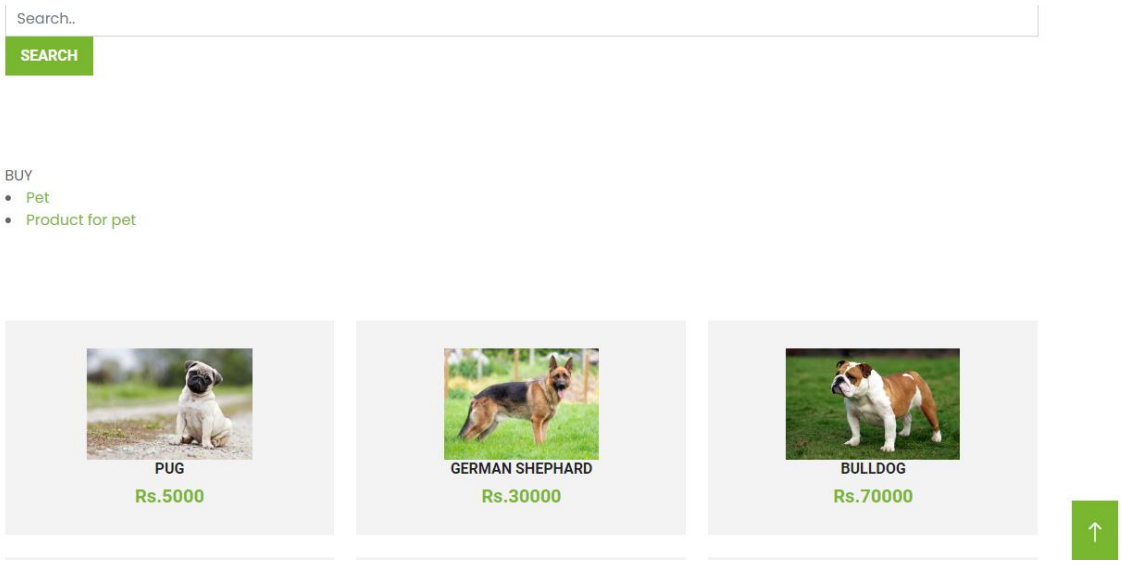

Figure 5.1.2 - 2: Screen Mock-up of Pet Store.

From home, when the user selects STORE on the navigation bar, this page is then displayed where the user can browse pets or products of their choice and add them to the cart. After that, they can place their order and proceed to checkout. While checking out, they can either pay instantly or pay “on delivery”. If they decide to pay instantly then they are provided with and softcopy receipt which they can print or download as they like.

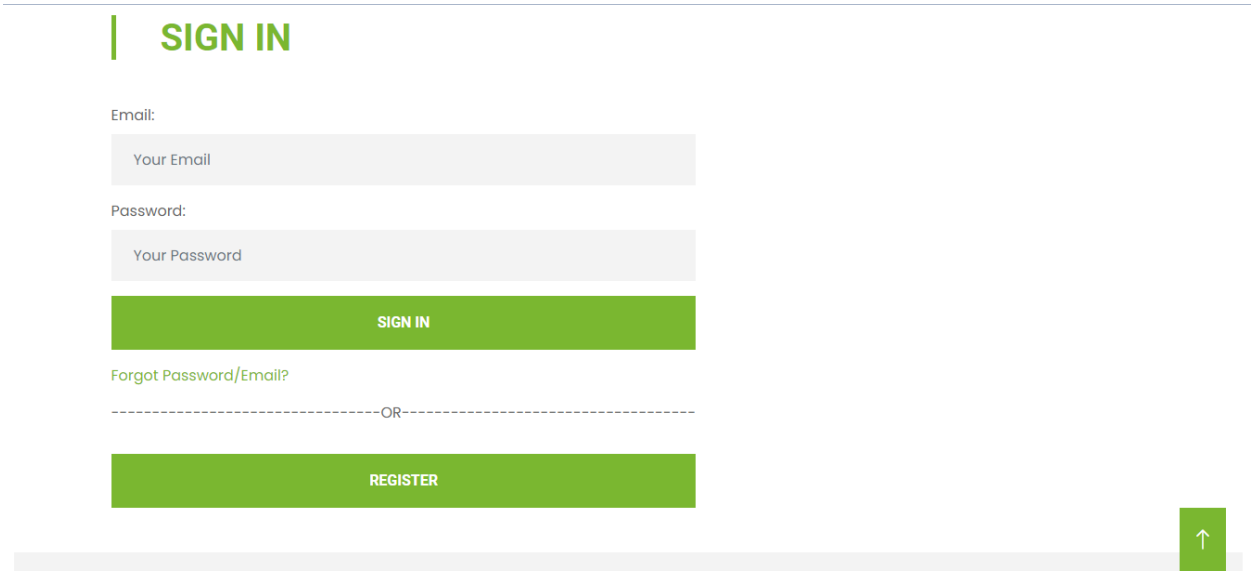

Figure 5.1.2 - 3: Screen Mock-up of Sign in page.

When a user selects MY ACCOUNT from the navigation bar, then this page is shown if they are not signed in, if they are signed in then USER PROFILE is displayed instead of this page. On this page, when a user enters the email and password and presses the sign-in button the system will check in database for the inputs. If there is data recorded in the database that the user had entered, then the system will allow the user to sign in and if there is no data recorded in the database then a message saying “Invalid email or password” will be displayed.

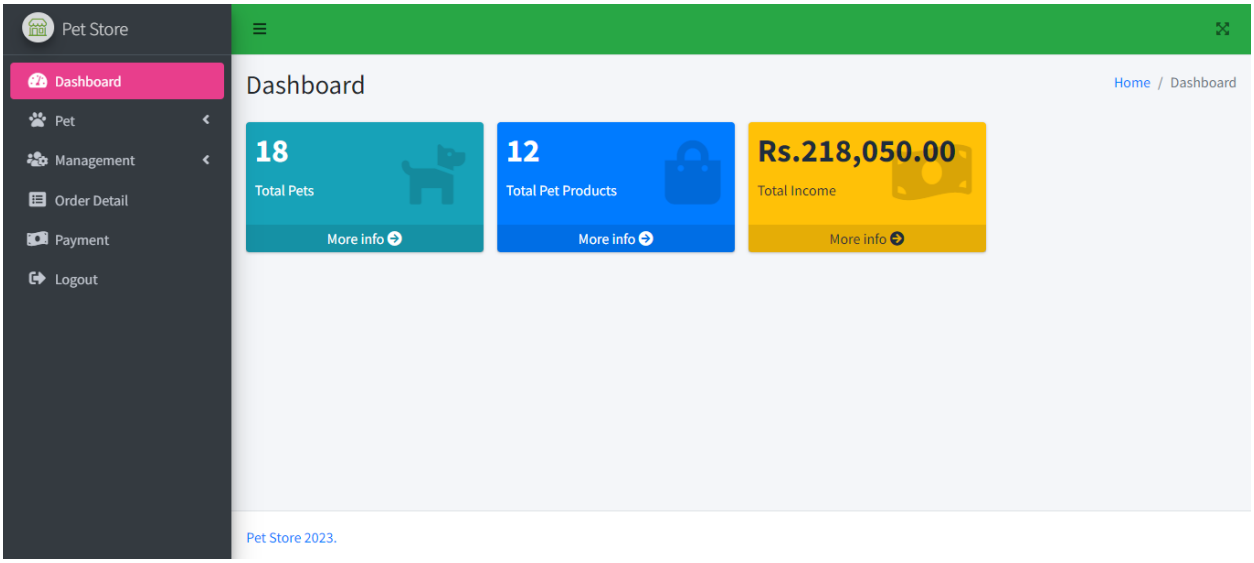

Figure 5.1.2 - 4: Screen Mock-up of Admin Dashboard(inettutor 2020).

This page is Admin Dashboard, it can be accessed by searching localhost/admin/ on the browser. Admins must sign in to get into this page. It displays the total number of pets, and products available in the system and calculates and displays total income. From here admin can navigate through different pages to manage pets, products, pet/product categories, orders, users, order details and payment.

6. System Build and Technical Notes

The design for the screen mock-ups was the first step in the software development process.
To develop the application, it was run on Docker Desktop (Docker Inc. 2023). The languages used to create the format of the webpages are HTML5 (W3C 2023) with the help of CSS (Håkon Wium Lie 2023) in styling and some alert messages, and functions with the help of Java Script (Brendan Eich 2023). For the website front-end icons, flat icon (Freepik Company 2023) was used and Bootstrap (Mark Otto and Jacob Thornton 2023) was integrated for the main structuring part of the coding. For the backend, plugins like jQuery (OpenJS Foundation 2023), and datatables (SpryMedia Ltd. 2023) were used and for the font, Font Awesome was chosen (Fonticons 2023).
The system is now linked to a database. By adding MySQL (Oracle 2023) instructions to the PHP code (The PHP Group 2023), the system can perform all CRUD operations (create, read, update, and delete); for an illustration, see Figure 6 - 1.

```php
<?php
session_start();
// <!-- this code section is for connecting the database -->
$hostname = 'db';
$username = 'admin1';
$password = 'password';
$dbname = 'pets';
$pdo = new PDO("mysql:host=$hostname;dbname=$dbname", $username, $password);

// for adding pet category ************************************************
if (isset($_POST['submit'])) {
  try {
    // Prepare the SQL statement
    $stmt = $pdo->prepare('INSERT INTO tbl_pet_category (pet_category_name, user_id) VALUES (:pet_category_name, :user_id)');

    $user_id_value = 1;

    // Bind parameters
    $stmt->bindParam(':pet_category_name', $_POST['pet_category_name']);
    $stmt->bindParam(':user_id', $user_id_value);

    // Execute the statement
    $stmt->execute();

    // Display success message
    echo "Pet category added successfully!";
  } catch (PDOException $e) {
    die("Error: " . $e->getMessage());
  }
}
```

Figure 6 - 1: Connection of MySQL database in PHP.

The database table for the program handles all operations that call for database access. Pages were made to facilitate the system's required events when it was determined what data was needed to fill the database tables.

With careful attention to the more subtle aspects when executing operations on numerous tables in the database, such as those found in the course, units, and assessments pages, many of the system's fundamental CRUD functions might be implemented using a similar code structure. To prevent errors, one must follow the constraints set by the database logic.

Search in store

A search functionality allows users to find specific information quickly. A search bar and results page are common components. The functionality to search starts with checking the submission form. An ‘if’ statement is implemented that checks if a POST request with a field name ‘search’ has been submitted then a search query is retrieved and assigned to the variable ‘$searchQuery’. Then a filtering query is entered to filter and select the entered word. A variable named $filteredPets holds the query to filter the searched word by the user. A variable named ‘$noResultsMessage’ is written to display a message that displays “no result found” if the search

word does not match the data in the database. At last, the function will be called in the html search form to work properly.

```php
//for search*****************************************************************************************
if (isset($_POST['search'])) {
    $searchQuery = $_POST['search'];

    // Filter the pets based on the search query
    $filteredPets = array_filter($pets, function ($pet) use ($searchQuery) {
        return stristr($pet['pet_name'], $searchQuery) !== false;
    });
    if (empty($filteredPets)) {
        $noResultsMessage = "No results found.";
    }
} else {
    // Display all pets by default
    $filteredPets = $pets;
}
?>
```

Figure 6 - 2: Code of Search function (CSK157 2022).

The code that will be called in the search button which will be displayed on the website is shown in Figure 6 – 3.

```html
<!-- Search Start -->
<div class="container py-5">
    <form method="POST" action="">
        <input class="form-control" id="myInput" type="text" name="search" placeholder="Search..">
        <button type="submit" class="btn btn-primary py-2 px-3">Search</button>
        <!-- <button class="btn">Advanced Search</button> -->
    </form>
</div>
<!-- Output even there are small or capital letters, when searching -->
<script>
    $(document).ready(function() {
        $("#myInput").on("keyup", function() {
            var value = $(this).val().toLowerCase();
            $("#myDIV *").filter(function() {
                $(this).toggle($(this).text().toLowerCase().indexOf(value) > -1)
            });
        });
    });
</script>
<!-- Search End -->
```

Figure 6 - 3: Search function called in html from.

User Registration

The code starts with an if statement that checks if a POST request with a field named "submit" has been submitted. This indicates that the user has submitted a registration form. If the form has been submitted, the code proceeds to retrieve various form input values from the $_POST super

global and assigns them to respective variables. These variables store user-provided data like the password, username, contact information, and email.

The code checks if the entered password matches the re-entered password ($repassword). If the passwords do not match, an alert is displayed to the user using JavaScript.

If the passwords match, the code proceeds to hash the password using the password_hash function. This is a security measure to store passwords securely in the database.

The program creates an SQL statement to add the user's information to the "tbl user" database table. To connect the data to the statement, it employs placeholders for username, password, re-password, contact information, and email. Additionally, it defines a $criteria associative array that associates the placeholders with the corresponding variables containing the user data.

Using the execute method of the prepared statement, the SQL statement is run along with the supplied data. An alert is displayed to let the user know a new account has been created if the insertion is successful. Additionally, JavaScript is used to route the user to a login page.

An alert is displayed to advise the user of the situation if an error happens while the SQL statement is being executed (for example, a database error).

```php
// registration start
if (isset($_POST['submit'])) {
    $password = $_POST['password'];
    $repassword = $_POST['repassword'];
    $username = $_POST['username'];
    $contact = $_POST['contact'];
    $email = $_POST['email'];

    // Check if passwords match
    if ($password !== $repassword) {
        echo '<script>alert("Passwords do not match.")</script>';
    } else {
        // Hash the password
        $hashedPassword = password_hash($password, PASSWORD_BCRYPT);

        $stmt = $pdo->prepare('INSERT INTO tbl_user (username, password, repassword, contact, email)
        VALUES (:username, :password, :repassword, :contact, :email)');
        $criteria = [
            ':username' => $username,
            ':password' => $hashedPassword,
            ':repassword' => $repassword, // Include the repassword field
            ':email' => $email,
            ':contact' => $contact,
        ];

        if ($stmt->execute($criteria)) {
            echo '<script>alert("New account created.")</script>';
            // Redirect to another page
            echo '<script>window.location.replace("login.php")</script>';
        } else {
            echo '<script>alert("An error occurred")</script>';
        }
```

Figure 6 - 4: PHP Code for Registration of User.

Image Integration

The program checks to see if data was sent using a form with a "submit" button. It searches for uploaded files, specifies a directory on the server where they will be stored, and then constructs the entire path to the file by combining the directory name with the file name. The code changes the file extension, such as.jpg or.pdf, to lowercase. It checks to see if the uploaded file's extension is on a list of permitted file extensions. If not, a warning is displayed.

It transfers the file from its temporary location to the designated directory if the file extension is permitted. This indicates that the file was successfully uploaded.

```php
// checking if a file was uploaded
if (isset($_FILES['pet_images']) && $_FILES['pet_images']['error'] === UPLOAD_ERR_OK) {
  // a directory to store uploaded files
  $upload_dir = '../../pimg/';
  $upload_file = $upload_dir . basename($_FILES['pet_images']['name']);

  // this code is to get the file extension
  $file_extension = strtolower(pathinfo($upload_file, PATHINFO_EXTENSION));

  // an array of allowed file extensions
  $allowed_extensions = array('jpg', 'jpeg', 'png', 'gif', 'pdf', 'doc', 'docx');

  if (in_array($file_extension, $allowed_extensions)) {
    // to move the uploaded file to the defined directory
    if (move_uploaded_file($_FILES['pet_images']['tmp_name'], $upload_file)) {
```

Figure 6 - 5: Code for Image upload (Tarun Singh 2023).

Under the following circumstances, this code inserts an error message into the $alertMessage variable:

if a file upload issue prevents the file from being moved to the desired directory.

The uploaded file does not match the permitted extensions if it has an incorrect type.

if no file was uploaded or if an uploading error took place.

In addition, a try...catch block is used in the code to manage any potential PDOException exceptions that may arise while doing database-related actions. If an exception is caught, the die is used to show an error message.

```php
        } else {
          $alertMessage = "Error uploading the file.";
        }
      } else {
        $alertMessage = "Invalid file type. Allowed types: JPG, JPEG, PNG, GIF, PDF, DOC, DOCX.";
      }
    } else {
      $alertMessage = "No file uploaded or an error occurred during upload.";
    }
  } catch (PDOException $e) {
    die("Error: " . $e->getMessage());
  }
}
```

Figure 6 - 6: Error/Exception handling.

During this period, the image was uploaded but was not displayed on the front end of the website, so I took help from (Coder Coder 2020) and solved my issue.

7. Test Strategy

A test strategy is implemented to guarantee that the finished product includes the anticipated capabilities specified in the requirements specification. Although the system is continuously tested as new features are added, future system development may lead to unforeseen development faults. As a result, a final strategy should be used to test the system once it has been completed and all functionalities have been included.

7.1. Overview of Test Strategy

Unit Testing:

To assess the separate units or components of a program, unit testing is a software testing approach used in software development. The smallest testable component of a software application, such as a function, method, or class, is referred to as a "unit" in the context of software development. Unit testing focuses on ensuring that these discrete pieces of code operate as anticipated and generate the right results for a given set of inputs.

The purpose of unit tests is to separate them from the rest of the software. Unit test creation, execution, and maintenance are made easier for developers by automated testing frameworks and tools. Unit tests ought to yield reliable outcomes. Since developers typically run unit tests while developing, it is expected that they will execute rapidly. Multiple unit tests may be run on a single unit to cover a range of possibilities and edge cases. The outcomes of other tests shouldn't be used in unit tests. They ought to be independent and not interfere with the results of other testing. Unit tests ought to produce predictable results. If a unit test fails, it ought to reveal a genuine bug in the code, and the error ought to be repeatable.

Unit testing reduces the cost of addressing errors, identifies flaws early in the development process, and gives developers the assurance that their code operates as expected, all of which contribute to higher software quality.

Black Box Testing:

Black box testing is a testing method that concentrates on testing a software application's functionality without considering its internal code or structure. Testing professionals treat the

software as a "black box," analysing its inputs and outputs to see if it performs as intended. Testers do not need to be familiar with the inner workings of the software because test cases are created based on the requirements and functionality of the product. This kind of testing can be used to find flaws, failures, or unexpected behaviour and is primarily concerned with verifying that the software performs as intended.

The functionalities are rated using these symbols:

'✓' a functionality that has been fully implemented.

'?' a functionality that has been partially implemented.

'*' is a functionality that has not been implemented.

Acceptance Testing:

The purpose of acceptance testing is to ascertain whether a software system satisfies the defined acceptance criteria and is prepared for deployment and use by the stakeholders or end-users. It is typically the last stage of the testing procedure and is used to make sure the software satisfies the needs of the user or the business. The two primary categories of acceptability testing are:

User Acceptance Testing (UAT) entails end users or end-user representatives running test cases to confirm that the software satisfies their needs and performs as expected in their real-world circumstances.

Acceptance testing ensures that the software satisfies the requirements and expectations of its intended users or stakeholders and is ready for production use.

These testing approaches are important in assuring the quality and dependability of software systems and they have diverse functions during the software development and testing process.

7.2. Sample Test Results

7.2.1. Unit Testing

Search Error

This is the error caused by a small typo mistake in my PHP code of the search function.

```
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
Deprecated: stristr(): Passing null to parameter #1 ($haystack) of type string is deprecated in /var/www/html/store.php on line 66
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
Deprecated: stristr(): Passing null to parameter #1 ($haystack) of type string is deprecated in /var/www/html/store.php on line 66
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
Deprecated: stristr(): Passing null to parameter #1 ($haystack) of type string is deprecated in /var/www/html/store.php on line 66
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
Deprecated: stristr(): Passing null to parameter #1 ($haystack) of type string is deprecated in /var/www/html/store.php on line 66
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
Deprecated: stristr(): Passing null to parameter #1 ($haystack) of type string is deprecated in /var/www/html/store.php on line 66
Warning: Undefined array key "petname" in /var/www/html/store.php on line 66
```

Figure 7.2.1 - 1: Error in the search function.

The attribute name is a pet name, which was supposed to be pet_name. This attribute is changed to check the fetching of data by pet_name and not by any other attribute names. The original code is in Figure 7.2.1 – 3.

```
    $searchQuery = $_POST['search'];

    // Filter the pets based on the search query
    $filteredPets = array_filter($pets, function ($pet) use ($searchQuery) {
        return stristr($pet['petname'], $searchQuery) !== false;
    });
    if (empty($filteredPets)) {
        $noResultsMessage = "No results found.";
    }
} else {
```

Figure 7.2.1 - 2: Changed attribute in the search function.

```
    $searchQuery = $_POST['search'];

    // Filter the pets based on the search query
    $filteredPets = array_filter($pets, function ($pet) use ($searchQuery) {
        return stristr($pet['pet_name'], $searchQuery) !== false;
    });
    if (empty($filteredPets)) {
        $noResultsMessage = "No results found.";
    }
```

Figure 7.2.1 - 3: Original code for a search function.

Login Error

For testing the login function, I missed a colon (:) so that it would be clear that the code retrieves an attribute name email in the user table.

Fatal error: Uncaught PDOException: SQLSTATE[HY093]: Invalid parameter number: number of bound variables does not match number of tokens in /var/www/html/login.php:23 Stack trace: #0 /var/www/html/login.php(23): PDOStatement->execute() #1 {main} thrown in **/var/www/html/login.php** on line **23**

Figure 7.2.1 - 4: Error in Login function.

Here this the code segment. The system cannot bind the email attribute with the input that the user provided here.

```
if (isset($_POST['submit'])) {
    // Get user input from the form
    $email = $_POST['email'];
    $password = $_POST['password'];

    // Prepare and execute a SQL query to fetch user data based on email
    $stmt = $pdo->prepare('SELECT user_id, email, password FROM tbl_user WHERE email = email');

    // Bind the email parameter
    $stmt->bindParam(':email', $email);
```

Figure 7.2.1 - 5: Missing colon in Login function.

```php
if (isset($_POST['submit'])) {
    // Get user input from the form
    $email = $_POST['email'];
    $password = $_POST['password'];

    // Prepare and execute a SQL query to fetch user data based on email
    $stmt = $pdo->prepare('SELECT user_id, email, password FROM tbl_user WHERE email = :email');

    // Bind the email parameter
    $stmt->bindParam(':email', $email);
```

Figure 7.2.1 - 6: Original code in Login function.

Add Pets Error

Adding the pet function shows an error saying that the file directory is not found, and the image is not able to move, which means that the image is not uploaded.

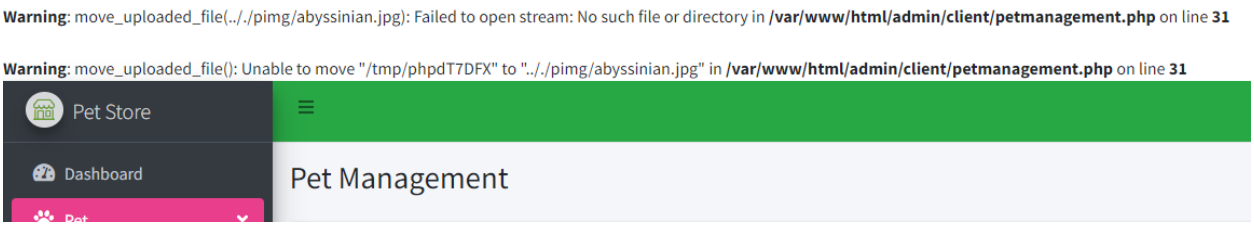

Figure 7.2.1 - 7: Error in adding pet function.

The error occurred because I changed the directory of the file path where the image was supposed to be uploaded, no such directory exists, so an error is occurring. In this case, I also missed a colon in the discount. The correct code is given in Figure 7.2.1 – 9.

```php
if (isset($_FILES['pet_images']) && $_FILES['pet_images']['error'] === UPLOAD_ERR_OK) {
  // a directory to store uploaded files
  $upload_dir = '.././pimg/';
  $upload_file = $upload_dir . basename($_FILES['pet_images']['name']);

  // this code is to get the file extension
  $file_extension = strtolower(pathinfo($upload_file, PATHINFO_EXTENSION));

  // an array of allowed file extensions
  $allowed_extensions = array('jpg', 'jpeg', 'png', 'gif', 'pdf', 'doc', 'docx');

  if (in_array($file_extension, $allowed_extensions)) {
    // to move the uploaded file to the defined directory
    if (move_uploaded_file($_FILES['pet_images']['tmp_name'], $upload_file)) {

      // Prepare the SQL statement
      $stmt = $pdo->prepare('INSERT INTO tbl_pet (pet_description, pet_name, pet_category_id, pet_images, pet_price, quantity, discount, us
                             VALUES (:pet_description, :pet_name, :pet_category_id, :pet_images, :pet_price, :quantity, discount, :user_id)'

      // Bind parameters
      $stmt->bindParam(':pet_description', $_POST['pet_description']);
```

Figure 7.2.1 - 8: Changed Directory in Add pet function.

```php
if (isset($_FILES['pet_images']) && $_FILES['pet_images']['error'] === UPLOAD_ERR_OK) {
  // a directory to store uploaded files
  $upload_dir = '../../pimg/';
  $upload_file = $upload_dir . basename($_FILES['pet_images']['name']);

  // this code is to get the file extension
  $file_extension = strtolower(pathinfo($upload_file, PATHINFO_EXTENSION));

  // an array of allowed file extensions
  $allowed_extensions = array('jpg', 'jpeg', 'png', 'gif', 'pdf', 'doc', 'docx');

  if (in_array($file_extension, $allowed_extensions)) {
    // to move the uploaded file to the defined directory
    if (move_uploaded_file($_FILES['pet_images']['tmp_name'], $upload_file)) {

      // Prepare the SQL statement
      $stmt = $pdo->prepare('INSERT INTO tbl_pet (pet_description, pet_name, pet_category_id, pet_images, pet_price, quantity, discount, us
                             VALUES (:pet_description, :pet_name, :pet_category_id, :pet_images, :pet_price, :quantity, :discount, :user_id)

      // Bind parameters
      $stmt->bindParam(':pet_description', $_POST['pet_description']);
```

Figure 7.2.1 - 9: Original code of Add pet function.

Empty Cart Error

This error is caused because the cart is empty, but the user is trying to place an order.

Your cart is empty. Please add items to your cart.
Warning: Undefined array key "cart" in **/var/www/html/checkout.php** on line **75**

Warning: foreach() argument must be of type array|object, null given in **/var/www/html/checkout.php** on line **75**

Warning: Cannot modify header information - headers already sent by (output started at /var/www/html/checkout.php:30) in **/var/www/html/checkout.php** on line **93**

Figure 7.2.1 - 10: Error Empty Cart.

I selected a pet in the cart, that is the cart is now not empty and the user can proceed to checkout and place an order.

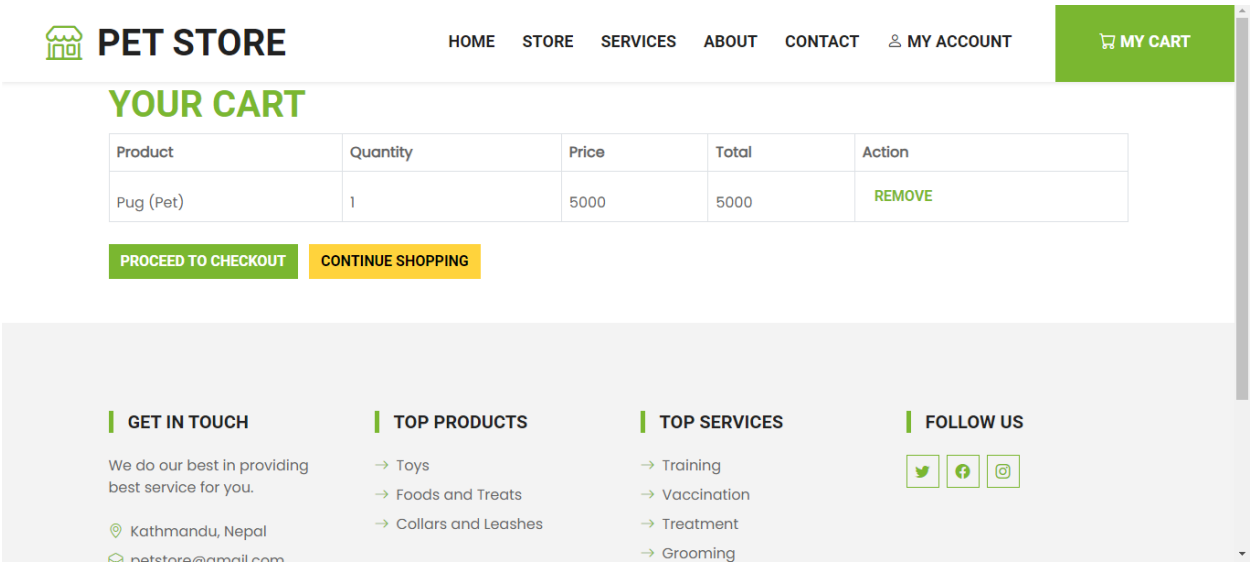

Figure 7.2.1 - 11: Proceed to checkout.

Proceed to the checkout button leads to this page where users can give their billing information and place an order.

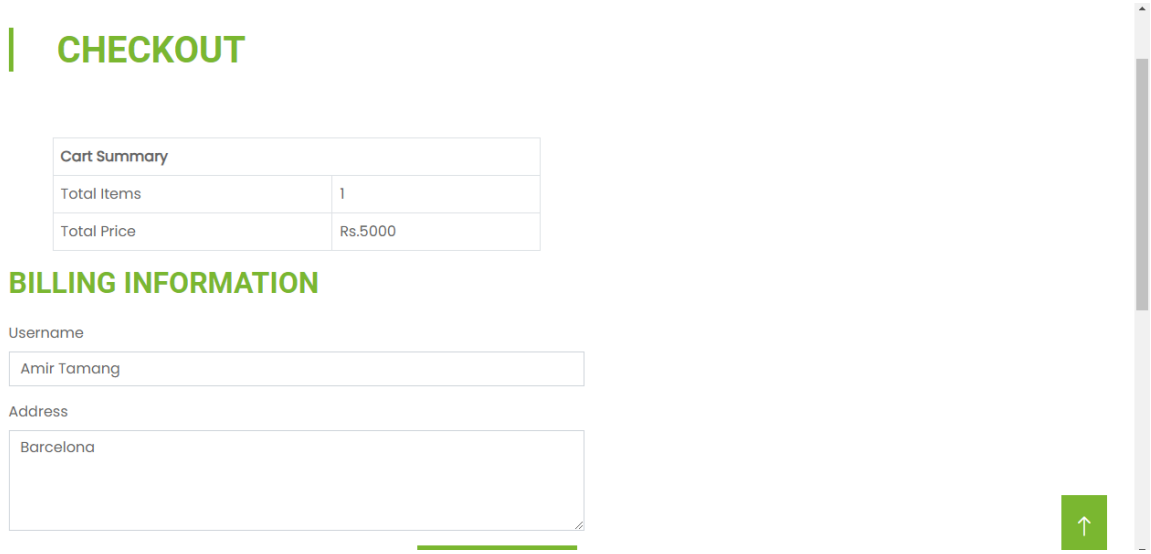

Figure 7.2.1 - 12: Checkout-Order.

7.2.2. Black Box Testing

Specification	**Achieved (✓ /? /*)**	**Comments**
Register	✓	Only customers can register, the admin user is already registered by default.
Login, Logout	✓	All registered users can log in and log in.
Login authentication	✓	When a user logs in with the wrong username or password it gives a flash message to enter the correct username or password.
Browse	✓	All users can browse through the website.

User Profile	✓	Every authenticated user has a profile with a default picture.
Search Field	✓	All the product modules have a search field. This search field gives results even when there is only some part of the name.
Add to cart	✓	Authenticated users can add any desired product to the cart, remove the products from the cart, and confirm the order.
Access to Profile, Order and Payment Pages	✓	Only authenticated users have access to the profile, orders and payment pages.
URL manipulation	✓	When a user manipulates the URL and changes the pet or product id. If the pet_id or product_id is not in the database, it redirects the user to the page not found error.
Admin Functionalities		
Admin Dashboard	✓	Dashboard in the admin panel to show the total number of

		pets, products and total income.
Pet and Product management	✓	Admin can add, edit and delete the pet and products from the database.
Pet and Product Categories Management	✓	Admin can add, edit and delete the pet and product categories from the database.
Order View & Delete	✓	Admin can view and delete the orders confirmed by the user.
View Order details and payment details.	✓	Admin can view the order details and payment details.
User and Order Management		Admin can manage any order or user as admin likes.

Table 7.2.2 - 1: Blackbox testing

7.2.3. Acceptance testing of the tool.

To showcase the product and gather feedback from the intended user group, testers were required to execute a list of tasks. As follows:

1. **Log into the website's admin panel using these details:**

User email: amir.yonjan12@gmail.com
Password: 123456

1a. Add a pet or product.

1b. Update the pet or product you just added.

1c. Delete the pet or product just updated.

1d. Log out of the admin panel.

2. Login to the website using these details:

Email: sid@gmail.com

Password: 654321

2a. Browse the store and add a pet or product to the cart.

2b. Proceed to checkout.

2c. Confirm the order.

2d. Pay (user only must put some value, real money will not be used.)

2e. View the receipt and Print/Save it as PDF.

2f. Log out from the website.

Results

Q1. Gender:

Male: 4

Female: 3

Tester use of the tool

Q2. Have you finished all the tasks listed on the task sheet?

Yes: 5

No: 2

Q3. How was the presented product?

4/5 (Acceptable to Excellent): 3

5/5 (Excellent): 4

Q4. Would your opinion of the tool affect how you felt about the pet store?

Yes: 6

No: 1

Q5. Do you consider a website like this for buying and selling pets to be a helpful tool?

4/5: 3

5/5: 4

Appearance and functionality

		Strongly disagree	**Disagree**	**Neither agree nor disagree**	**Agree**	**Strongly Agree**
Q6.	The on-screen layout of the website is easy to navigate					7
Q7.	Information is stored in a logical location in the system.				5	2
Q8.	I immediately understood the function of each menu item.			3	2	2
Q9.	I like the system layouts and design.				3	4
Q10.	The payment was easier for me		2	4	1	

Table 7.2.3 - 1: Appearance and function test.

Future improvements

Q11. Would you like to see the tool used on the website for the pet store with additional features?

Yes: 7

No: 0

Q12. Did the system present any issues for you?

a. No problems at all.
b. Nope.
c. It has some issues with the payment system.
d. No problems.
e. In the admin panel when I reloaded the page, the same pet was added which I added recently.
f. Not logging me in.
g. No.

Q13. Rate these features (1 is the least and 5 is the most):

Functionalities	**1**	**2**	**3**	**4**	**5**
Adding Pet / Products / Categories					7
Deleting Pet / Products / Categories					7
Editing Pet / Products / Categories			1	2	4
Search in store			2	3	2
Profile				3	4
User management					7
Order Management		2	3	2	
Payment Viewing			3	4	
Dashboard					7

Table 7.2.3 - 2: Functionality rating results.

Q14. Any other comments?

a. The website is easy to use and has a beautiful appearance. The supplies I needed for my pets were simple to find, and the payment process went without a hitch." When I used the "Find a Store Near Me" tool, it correctly located the neighbourhood pet store. This function helps me organise my visits."

b. I value detailed product descriptions and pictures. It made it simple for me to choose the right pet food and accessories.
c. The website is responsive and well-optimized, and I was able to access it from my mobile device. It was simple to shop and browse on my phone.
d. It's convenient to manage my pet profiles and keep track of previous purchases in the 'My Account' section. It allows me to quickly reorder products.
e. I put the search tool through a lot of testing, and it constantly produced useful results. Users must be able to easily locate pet goods.

8. System Evaluation

The system evaluation must determine if the product has solved the problems.
The final product has been presented to the target user group (Computing Students at NAMI College) to trail the system by conducting tasks and giving their feedback.
Following this, the final product was presented to the client. During an interview with Mr. Khem Raj Sharma.

- Mr. Sharma trialled the product to ascertain functionality, to assess that the appropriate level of genericity has been achieved,
- Feedback from the system testing and user-group trails will be given,
- This involved a review of the project aim and objectives regarding client satisfaction before Mr. Sharma decided if the store would be implementing the new system.

8.1. Usability Evaluation

Login:

The login functionality is straightforward to use. Users can easily access their accounts with their credentials. It's user-friendly and doesn't present any barriers to entry.

Pet/Product Search:

The search feature is efficient and effective. Users can quickly find pets or products they are looking for by entering relevant keywords or filters. The search results are relevant and displayed clearly.

Adding Pet/Product/Categories:

Adding pets, products, and categories is intuitive and well-organized. The process guides users through adding information step by step, making it simple for both novice and experienced users. Clear labelling and options contribute to a positive user experience.

User Management:

User management is well-implemented. Administrators can easily create, modify, or delete user accounts. Permissions and roles are clearly defined, ensuring that the right users have the appropriate access levels.

Order Management:

The order management functionality is efficient. Administrators can view and manage orders, update order statuses, and track deliveries easily. The interface is well-structured and provides essential order details briefly.

Order Details:

Accessing order details is straightforward for users. They can view comprehensive information about their orders, including order status, items, shipping details, and payment information. This transparency enhances user trust and satisfaction.

Payment:

Usability: The payment process is seamless and secure. Users have multiple payment options, and the checkout flow is intuitive. Clear instructions are provided, and users receive confirmation of successful payments.

General Issues:

Payment method needs to be made, currently, payment methods are not secure, and any payment services are not integrated.

Users can easily navigate, search for pets/products, manage their accounts, place orders, and make payments without encountering significant usability issues. This contributes to a positive user experience and customer satisfaction. However, it's important to continuously monitor and gather user feedback to make iterative improvements and address any potential usability issues that may arise in the future.

8.2. Project Conclusion/ Review

My main goal in designing and implementing my pet store website project was to build a platform that is user-centric and easily links pet lovers with the animals and goods they love. We have succeeded in attaining this goal, according to the usability assessment of major capabilities.

The website's key features, including Login, Pet/Product Search, Adding Pet/Product/Categories, User Management, Order Management, Order Details, and Payment, have all been carefully planned out and put through their paces. The usability assessment's findings demonstrate the superb user experience this website offers. The website's login mechanism provides a simple way for users to access their accounts, making it easy for them to do so. Users may quickly and easily discover the dogs or things they want with the help of the Pet/Product Search feature, which improves their browsing experience. The task of content maintenance is made simpler by the simple and intuitive Add Pet/Product/Categories approach. Strong functionalities for user management and order management enable administrators to easily manage user accounts and orders. To learn more about an order, the admin can read the order details. Finally, the payment system provides a checkout process to ease user-to-user transactions.

The website is now both functional and aesthetically pleasing because I adhered to best practices in user interface design, clear labelling, and effective navigation throughout the development process. I created a platform that is inclusive and accessible by taking the preferences and needs of both new and seasoned users into account.

The usability assessment confirms our conviction that the pet store website not only serves as an excellent resource for pet lovers but also serves as a symbol of our commitment to providing a top-notch online experience.

I am committed to upholding the high standards I have established and looking for possibilities for growth as I complete this project. To ensure that the pet store website continues to develop and please the users, user feedback and periodic usability assessments will remain crucial to our efforts.

Finally, my project shows the fruition of hard work, originality, and a user-centric perspective. I'm pleased to show off a pet supply store website that not only meets but also surpasses user expectations. I see a future characterised by expansion, user pleasure, and a long-lasting good influence on the pet enthusiast community thanks to the strong base already in place.

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Appendix 1 – Project time scales

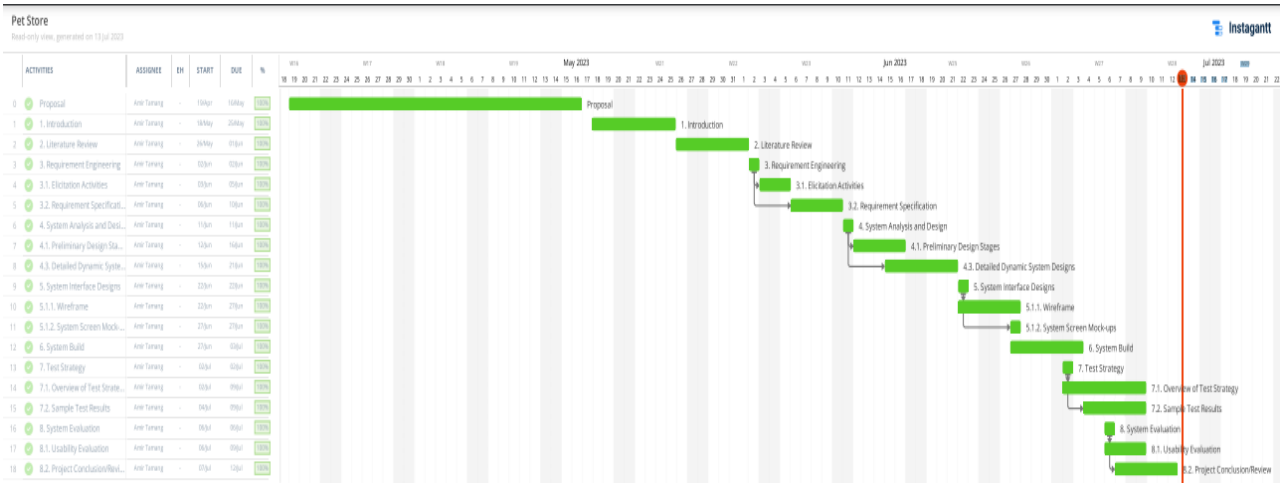

Figure Appendix 1 - 1: Original Project Time-Scales (Instagantt 2023).

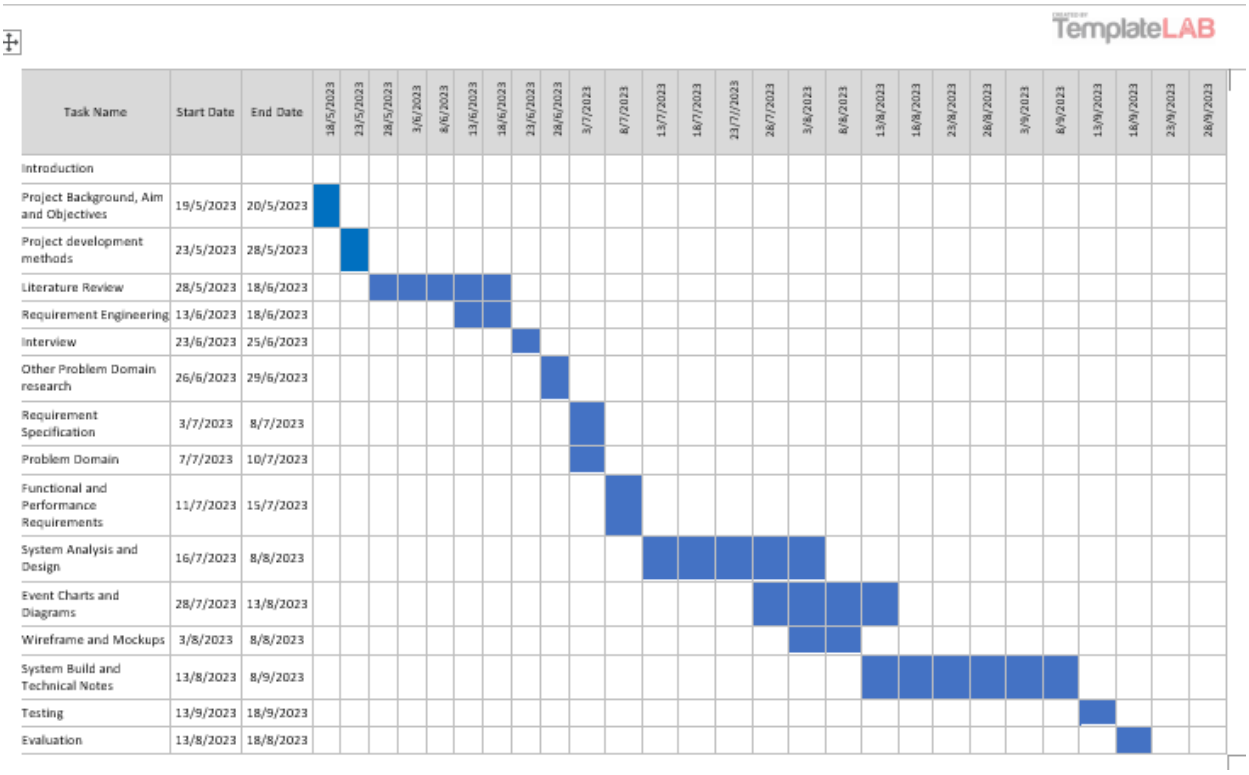

Figure Appendix 1 - 2: Updated Project Time-Scales (TemplateLAB 2020).

Appendix 2 - Project Diary Logs

PROJECT DIARY/LOG 1

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 14 April 2023 **Last diary date:**

Objectives:

- Research topic for dissertation: Pet buy, sell and adopt website.
- Similar system existence to understand how pet services are done in Nepal

Progress:

- Need to provide similar system by next meeting
- Complete understanding of the dissertation requirements

Supervisor's Comments:

The student project seems flexible and complete the objectives of the module requirement through report and practical approach.

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PROJECT DIARY/LOG 2

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 5 June 2023 **Last diary date:**

Objectives:

- Gather more research papers for literature review which will be used in proposal report.
- Final report of proposal report
- Understand development requirement.

Progress:

- Research topic for dissertation is chosen which is Pet buy, sell and adopt website. Gathered research papers for literature review and proposal writing.
- Show the system development progress by next meeting

Supervisor's Comments:

The student has completely understand development cycle with methodology and drafted version proposal shown.

PROJECT DIARY/LOG 3

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 14th June 2023 **Last diary date:**

Objectives:

- Present progress made in the interim report
- Implement the designed system through wireframe and front page

Progress:

- Complete the interim report and send for final report as soon as possible
- Encourage to complete the development of websites

Supervisor's Comments:

The student seems good progress with interim report and development.

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PROJECT DIARY/LOG 4

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 27th July 2023 **Last diary date:**

Objectives: Complete the product functionalities.

- Complete all the required functions of the website.
- Functionalities like add, edit, delete, display.

Progress:

- Presented progress made in the report.
- Presented and implemented a designed system through the wireframe and mock-ups.

Supervisor's Comments:

Progress is made in the report. Also implemented a designed system through the wireframe and mock-ups. Proceed to the next step.

PROJECT DIARY/LOG 5

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 10th September 2023 **Last diary date:**

Objectives:

- Complete the rest of the functionalities of the project which are order details, payment gateway and search bar.
- Display image in front-end which was which was uploaded in back-end (not displayed).
- Start testing and evaluation of the product.

Progress:

- Completed 80% of the required functions of the website.
- Technical build and notes of the project report are completed.

Supervisor's Comments:

Almost all of the required functions of the website are completed and technical build and notes of the project report are completed. Keep it up.

PROJECT DIARY/LOG 6

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 17th September 2023 **Last diary date:**

Objectives:

- Complete the testing and Evaluation in the project report.
- Focus on enhancing report quality.
- Prepare for Presentation and Viva.

Progress:

- Completed the rest of the functionalities of the project which are order details, payment gateway and search bar.
- Uploaded images from the backend displayed in the front end.
- Completed testing using two different methods of the product.

Supervisor's Comments:

All the required functionalities are completed. Now focus on completing the report and its quality.

PROJECT DIARY/LOG 7

Student: Amir Tamang **Supervisor: Ravi Chandra Gurung**

Date: 22nd September 2023 **Last diary date:**

Objectives:

- Submit project and report in NILE.

Progress:

- Completed the testing and Evaluation in the project report.
- Enhanced the report quality.
- Presentation and viva done.
- Completed project report.

Supervisor's Comments:

All work for the dissertation is completed and the report quality is very good. You may now submit your work.

Appendix 3 - Questionnaire

For Website Administrators/Owners:

1. What were the primary goals and objectives of launching the pet store website?

   -

2. Have there been any significant technical challenges or issues with the website? If so, how were they addressed?

   -

3. How do you handle customer complaints or disputes?

   -

4. Are there plans for expanding or improving the website in the future? If yes, what are they?

   -

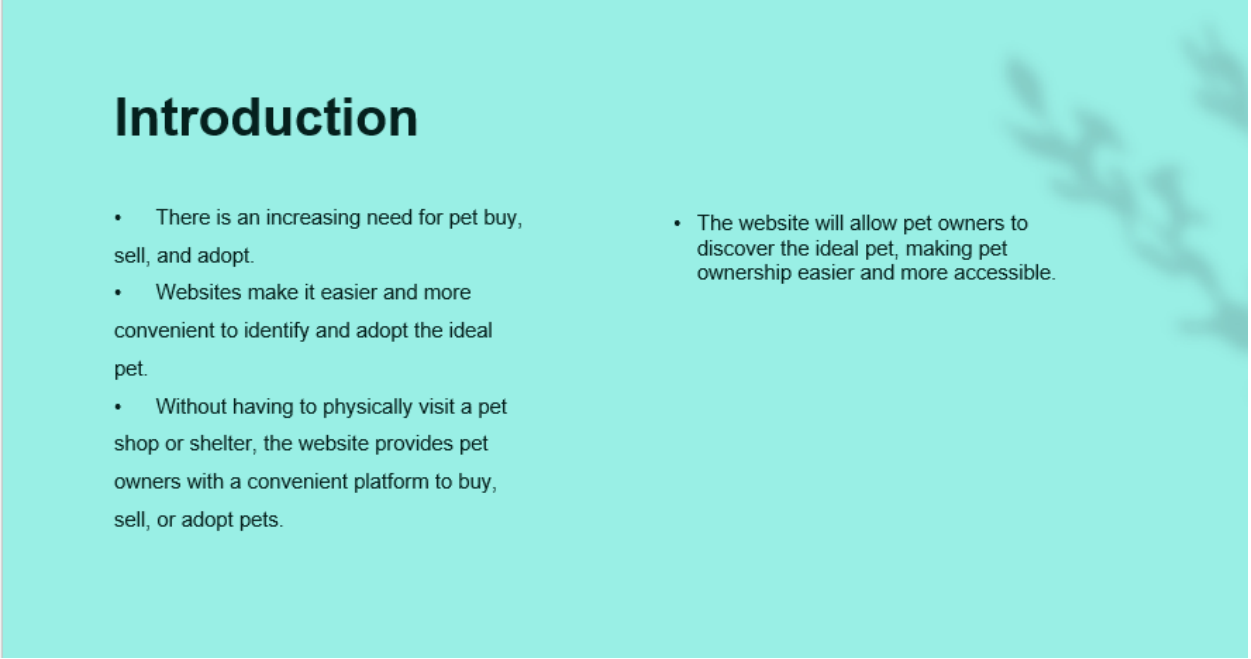
5. What is your vision for the future of the pet store website, and how do you plan to achieve it?

   -

6. How do you stay updated on industry trends and best practices to maintain the website's competitiveness?

   -

Appendix 4 - Presentation Slides



Aims & Objectives

- Develop a user-friendly website for purchasing and selling pets.
- Simplify the process of pet transactions and provide a convenient and reliable platform for pet enthusiasts.
- Facilitate responsible pet ownership, connect pet lovers, and contribute to the well-being and happiness of pets and their owners.

Problems

- Trust and Safety Concerns: Users often face trust issues when buying or selling pets or pet products online.
- Scams and Fraud: Both buyers and sellers can fall victim to scams, such as fake listings, nonexistent pets or products, and fraudulent payment schemes, eroding trust in the marketplace.
- Quality Assurance: Buyers may be unsure about the quality or authenticity of pet products they find online, raising concerns about the health and safety of their pets.

Solutions

- Secure Payment Options: Offer secure payment methods to protect users from fraudulent transactions.
- Customer Support: Offer responsive customer support to address user concerns and disputes promptly.
- Transparency: Clearly communicate platform operations, fees, and policies to build trust with users.
- Community Guidelines: Develop and enforce guidelines to promote responsible pet ownership and ethical conduct among users.

Methodology

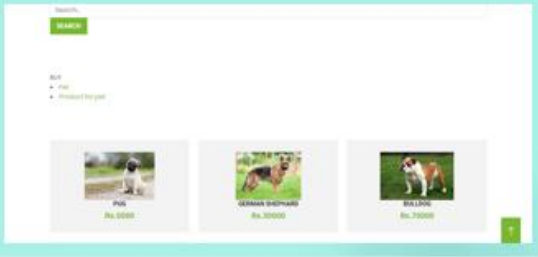
- Methods include in-depth interviews and focus groups.
- Methods involve surveys and data analysis.
- To gather insights from potential clients, pet organizations, and local pet enthusiasts.
- Agile development with iterative testing and user feedback.
- Work is in the Agile(scrum), focusing on different platform features.

Product Overview

- Offers secure payment methods to protect users from fraudulent transactions.
- Offers responsive customer support to address user concerns and disputes promptly.
- Clearly communicate platform operations, fees, and policies to build trust with users.
- Develop and enforce guidelines to promote responsible pet ownership and ethical conduct among users.

Wireframe

Screen Mockup



Product Benefits

- Secure payment method is done which helps users to protect their data.
- Customer support is very much responsive as the priority is to clear customer's concerns.
- Trustable to users.
- There are guidelines to promote responsible pet ownership and ethical conduct among users.

Development Required

- Web technologies include PHP, Bootstrap4, HTML, CSS, JavaScript.
- Plugins include Flat icons and jQuery.
- MySQL database
- Visual Studio Code is used as the database browser for the Laravel framework and Docker.

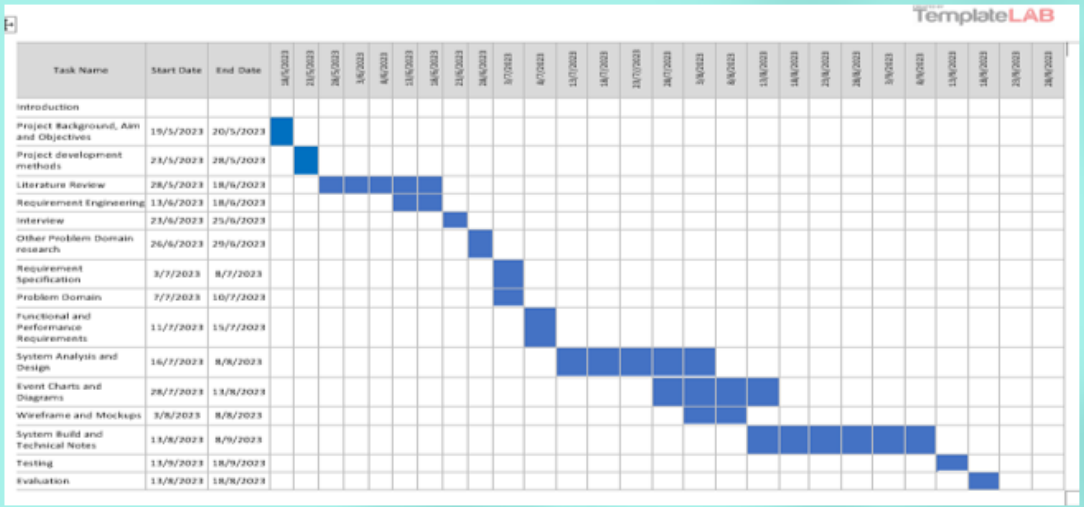
Summary

- The website allows pet owners to discover the ideal pet.
- This website makes it easier and more convenient to identify and adopt the ideal pet.
- People can shop online while sitting in their comfort zone.
- Secure payment.
- Customer support is responsive.
- Guidelines available to promote responsible pet ownership and ethical conduct among users.

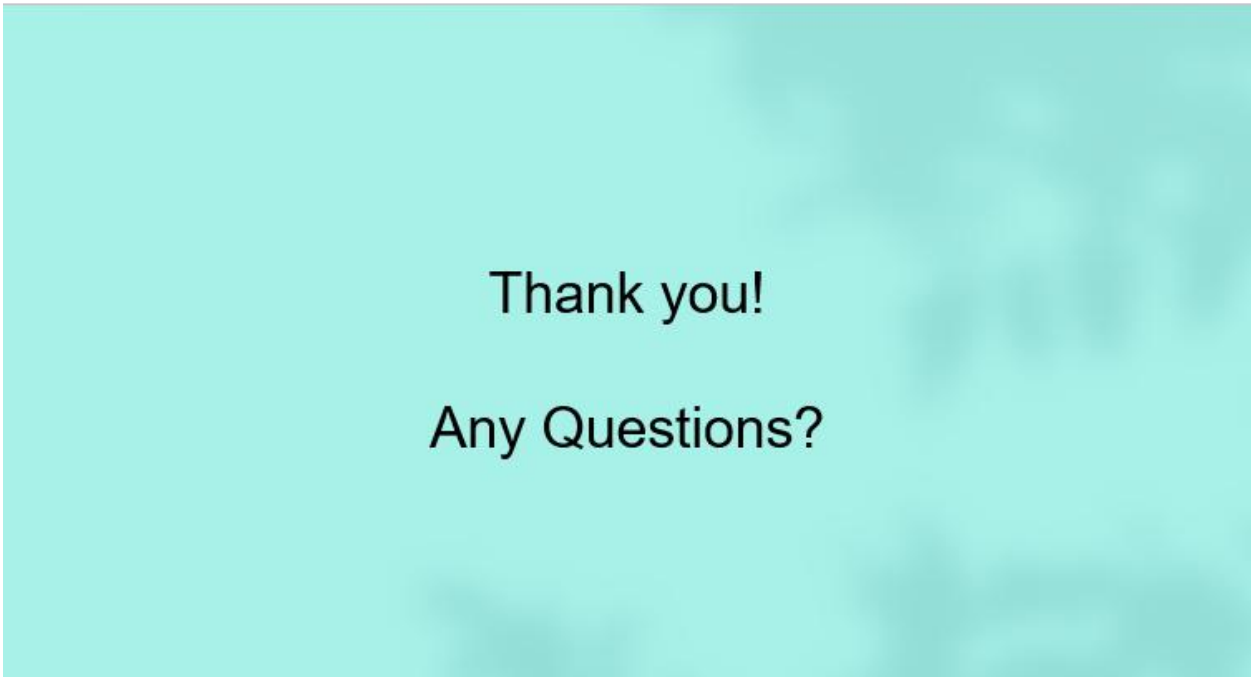
Future Scope

- Pet Adoption
- Advanced search filters
- Service management
- AI chatbots

Final Time-Scale



Task Name	Start Date	End Date
Introduction		
Project Background, Aim and Objectives	19/5/2023	20/5/2023
Project development methods	21/5/2023	28/5/2023
Literature Review	28/5/2023	18/6/2023
Requirement Engineering	13/6/2023	18/6/2023
Interview	23/6/2023	25/6/2023
Other Problem Domain research	26/6/2023	29/6/2023
Requirement Specification	3/7/2023	8/7/2023
Problem Domain	7/7/2023	10/7/2023
Functional and Performance Requirements	11/7/2023	15/7/2023
System Analysis and Design	16/7/2023	8/8/2023
Event Charts and Diagrams	28/7/2023	13/8/2023
Wireframe and Mockups	3/8/2023	8/8/2023
System Build and Technical Notes	13/8/2023	8/9/2023
Testing	13/9/2023	18/9/2023
Evaluation	13/8/2023	18/8/2023


Thank you!
Any Questions?